

## Research Report on

# Machine-learning models linking earnings call text and audio to short-horizon stock volatility

Full search query: I want to find research on predicting very short-horizon stock return volatility—covering both short-term and event-window measures, and including realized and implied volatility—using machine learning and deep learning models that exploit both textual and audio features from earnings conference calls, in either cross-sectional or time-series prediction setups

## Summary

The literature provides a fairly mature line of multimodal (text + audio) ML/DL models that forecast short horizon realized volatility over 1–30 trading days after earnings calls—anchored around 3 day post call volatility—but essentially no work yet on intraday/event window realized volatility or options implied volatility aligned precisely to call timing [1,2,3,5,6,7,9,10,11,16,17,18,19].

### Overall state of the field relative to your goal

#### Key takeaway

- **What exists:** There is a rich, growing body of work that predicts short horizon realized volatility from earnings conference calls using joint text–audio models, mostly at 3–30 day horizons and using daily returns [1,2,3,4,5,6,7,8,9,10,11,13,15,16,17,18,19]. Architecturally, the field has progressed from RNNs with hand crafted prosodic features [1] to hierarchical Transformers [2], graph based and network aware models [3,14], numeral aware LMs [15,16,17], LLM augmented pipelines [6,9,10,12], and physics informed acoustic/affective modeling [11].
- **What is missing:** Almost no paper tackles:
  - Truly “very short horizon” volatility in your strict sense (intraday, call start–close windows, or explicitly defined [ 1,+1] event windows, or next session intraday RV), and
  - Options implied volatility (or IV changes) as a primary predictive target. All major multimodal papers derive volatility from daily price data over fixed n day windows post call [1,2,3,5,6,7,9,10,11,15,16,17,18,19,20], and none explicitly build IV surfaces around calls.

This means your project can draw heavily on existing multimodal architectures and datasets, but the particular combination “intraday/event window realized volatility or IV, aligned to call timing, predicted from text+audio” is effectively open.

## 1. Core multimodal benchmarks for short horizon realized volatility

These are the most central references for ML/DL volatility prediction from earnings call text and audio.

### 1.1 Foundational multimodal models (RNNs, Transformers, GCNs)

- **Qin & Yang (ACL 2019, MDRM) [1]**
  - First widely used multimodal volatility model with aligned text + audio:
    - Text: sentence level GloVe embeddings.
    - Audio: 27 Praat vocal features (pitch, intensity, jitter, HNR, etc.) aligned to sentences.
  - Target: standard deviation / realized volatility based on returns over 3, 7, 15, 30 trading days after the call.
  - Design: RNN based sequence model with multimodal deep regression (MDRM).
  - Result: substantial gains over past volatility and text only baselines, with the strongest improvements at 3 day horizons.

- **HTML (The Web Conf 2020) [2]**
  - Upgrades MDRM to a **hierarchical Transformer**:
    - Text: sentence level Transformer encoder.
    - Audio: same style of 27 Praat features, sentence aligned.
    - Multimodal fusion layer + multi task head.
  - Targets: log volatility over n day post call windows (n=3,7,15,30), jointly predicting:
    - Average n day log vol.
    - Single day log vol.
  - Result: 17–49% error reductions vs baselines, especially for 3 day volatility. HTML is now a standard baseline.
- **VoITAGE (EMNLP 2020) [3]**
  - Adds two important elements:
    - Text encoder: FinBERT for sentence representations.
    - Multimodal fusion: inter modal attention to model verbal–vocal coherence.
    - Stock level graph: GCN over a stock correlation network.
  - Targets: average log volatility over a multi day window  $v[0,n]$  and single day log volatility  $v_n$  (n not explicit in the excerpt but consistent with 3–30 days).
  - Result: Outperforms MDRM and HTML, showing value of both cross modal attention and stock network structure.

Collectively, [1,2,3] define the canonical setup:

- Input: sentence aligned transcript + audio features.
- Horizon: 3–30 trading days after the call; realized volatility from daily returns.
- Model: sequence encoders (LSTM/Transformer) plus fusion; sometimes network structure.

## 1.2 Dataset and multi task extensions

- **MAEC dataset (CIKM 2020) [13]**
  - Large scale, aligned multimodal ECC dataset: 3,443 calls from 1,213 S&P 1500 firms (2015–2018).
  - Provides sentence level alignment, linguistic features, acoustic features, and MFCCs.
  - Includes volatility prediction baselines, but detailed label construction is not fully specified in the excerpt.
  - Importance: becomes the main resource for subsequent multimodal models (NumHTML [15], AMA LSTM [7], DocFin [8]).
- **Multimodal Multi Task Financial Risk Forecasting (ACM MM 2020) [4]**
  - Jointly predicts volatility and price movement from text + audio using a multimodal attention model.
  - Reinforces volatility and direction as complementary tasks for risk modeling.
- **NumHTML (AAAI 2022) [15]**
  - Extends HTML with:
    - Numeric oriented hierarchical Transformers (special handling of numerals).
    - Multi task learning: return prediction as main task; volatility as auxiliary.
  - Still uses multimodal text + audio from MAEC style data.
  - Shows that better numeral handling in text improves both returns and volatility prediction.

These works give you:

- Standard data (MAEC) and benchmarks.
- Multi task formulations that couple volatility with returns or other risk metrics.

## 2. Modern enhancements: LLMs, richer audio, and multi source fusion

Recent work makes two major leaps: leveraging large language models for text and more sophisticated representations for audio and other data sources.

### 2.1 LLM centric ECC analyzers and multi source risk systems

- **ECC Analyzer (AIFinance 2024 + arXiv) [6,12]**
  - Pipeline:

- Use a large language model to:
  - Summarize paragraphs of the transcript.
  - Perform RAG over the transcript using a finance specific "Question Bank" to extract salient sentences.
- Extract embeddings:
  - Text: LLM derived embeddings of summaries and key sentences.
  - Audio: embeddings from pretrained audio models (no longer just Praat features).
- Fuse embeddings into a regression model to predict n day volatility.
- Targets: multi horizon n day volatility (3,7,15,30 days), realized from daily prices.
- Evaluation: temporal split (80/20), benchmarks include GARCH, LSTM, MDRM, HTML, and a GPT 4 baseline.
- Result: clear improvements on 3 and 7 day horizons; comparable performance at 15–30 days.
- **RiskLabs (2024, arXiv + journal) [9,10]**
  - Comprehensive multimodal, multi source risk framework:
    - Modalities:
      - Earnings call: audio + transcripts (LLM derived text embeddings + acoustic embeddings).
      - Pre call time series: 30 days of historical prices.
      - Firm centered daily news around ECC dates.
    - Architecture: separate encoders (Earnings Call Encoder, Time Series Encoder, News Market Encoder) and fusion layers.
    - Tasks: multi task prediction of volatility/variance and 95% VaR over 3,7,15,30 days.
  - Key insights:
    - LLMs are best used as feature extractors, not standalone forecasters.
    - Combining call content with prior price dynamics and news yields sizable gains.

For your work, these papers show:

- How to integrate LLM based semantic extraction with audio and other information sources.
- A practical blueprint for multi source, multi task volatility forecasting systems anchored on earnings calls.

## 2.2 Physics informed audio and affective state modeling

- **The Sound of Risk (2025) [11]**
  - Focus: robust and interpretable acoustic modeling.
  - Audio:
    - Uses a Physics Informed Acoustic Model (PIAM) to extract paralinguistic emotional cues (tension, stability, arousal) robustly across noisy call environments.
    - Validates PIAM on DNS benchmarks.
  - Text:
    - LLM derived textual sentiment/features mapped into the same 3 D Affective State Label (ASL) space (Tension, Stability, Arousal).
  - Features:
    - ~150 call level ASL features per executive and section (presentation vs Q&A), including:
      - Moments of ASL, within call dynamics, and differences between presentation and Q&A.
    - Financial controls such as 30 day historical volatility.
  - Targets:
    - Realized volatility over 1, 7, and 30 days post call.
    - Cumulative abnormal returns (CAR).
  - Results:
    - Multimodal affective features explain up to ~44% of out of sample variance in 30 day volatility; strong predictive contributions from affective shifts in Q&A, especially CFO instability.
    - Little predictive power for CAR, highlighting a divergence between volatility/risk and directional performance.

Relative to your "very short horizon" interest, [11] is particularly important because:

- It includes an explicit 1 day realized volatility target (closest to your horizon).
- It demonstrates how subtle affective dynamics in speech (not just average prosody) carry strong volatility information.

### 3. Specialized themes: numerals, dialogue structure, denoising, and fairness

Even when not all are multimodal, these streams directly shape how multimodal models are designed and evaluated.

#### 3.1 Numeral understanding and short term volatility

- **ECNum + NAM (CIKM 2021) [17]**
  - Introduces ECNum, annotating numeral types (monetary amounts, percentages, dates, etc.) in ECC transcripts.
  - Proposes a Numeral Aware Model (NAM) that explicitly captures numeral semantics.
  - Demonstrates improved short term volatility forecasting (3–7 day horizons) relative to MDRM and HTML.
- **GNA Vol (IJCNLP 2023) [16]**
  - Proposes a pretraining task (General Numeral Attachment) to help LMs correctly attach numerals to entities in text.
  - Shows this pretraining significantly improves 3 and 7 day volatility prediction versus strong baselines including multimodal MDRM and HTML.

Implication:

- For very short horizon RV or IV (where specific numbers—EPS surprises, guidance, margin levels, option strikes—are critical), numeracy focused pretraining and architectures from [16,17] should be directly exploitable.

#### 3.2 Dialogue/Q&A structure and text denoising

- **Multi Round Q&A Attention Network (IJCAI 2020) [18]**
  - Uses a dialogue aware attention network over multi round Q&A structure in transcripts.
  - Shows that modeling the alternation and structure of Q&A improves volatility forecasts over several periods, beyond document level features.
- **DeFVP (ICME 2024) [19]**
  - Treats time series (price/volatility) and text as two modalities.
  - Introduces a Differentiable Binary Selector (DBS) to decide, sentence by sentence, which text segments should influence the volatility predictor (VP).
  - Demonstrates that learned sentence selection significantly improves volatility prediction on multiple datasets, highlighting that not all ECC text is informative.

Implication:

- For your high frequency/event window setting, where only parts of the call may trigger significant volatility changes, dialogue aware and denoising/selection methods like [18,19] could be especially valuable.

#### 3.3 Fairness, bias, and robustness

- **Empirical Investigation of Bias in Multimodal ECC Analysis (NAACL 2021) [5]**
  - Uses multimodal volatility models on S&P 500 calls (volatility labels  $v[0, \tilde{A}]$ ,  $\tilde{A}=3,7,15,30$ ).
  - Splits data by management speaker gender and shows:
    - Multimodal models often outperform text only on average.
    - But error disparities across male vs female executives can increase, indicating gender bias amplified by audio features.
- **DocFin (EMNLP 2022) [8]**
  - Fuses text, audio, and semi structured tabular data (10 K/10 Q tables).
  - Shows:
    - 5–12% improvements in volatility and price movement prediction vs audio+text alone.
    - Over 30% reduction in gender related prediction bias by adding tabular data.
- **AMA LSTM (2024) [7]**
  - Adversarially trained multimodal LSTM using FinBERT text embeddings and audio features.
  - Aims at volatility prediction while:
    - Injecting adversarial perturbations to simulate market noise.
    - Actively reducing gender bias in audio driven predictions.

- Finds audio carries more gender bias than text, but multimodal fusion and adversarial training can improve both robustness and fairness.

Implication:

- Any future model that uses vocal cues to predict very short horizon volatility or IV should integrate fairness diagnostics and possibly adversarial or multi source mitigation strategies along the lines of [5,7,8].

## 4. What is and is not covered relative to your precise volatility target

### 4.1 Horizons and realized volatility construction

- Almost all multimodal ECC studies:
  - Use realized volatility built from daily close to close returns over fixed n day windows post call (n=1,3,7,15,30) [1,2,3,5,6,7,9,10,11,15,16,17,18,19,20].
  - Do not:
    - Distinguish pre market vs post market call timing for label alignment.
    - Construct intraday realized variance (e.g., 5 minute RV).
    - Explicitly define event windows such as call start–market close or [ 1,+1] days with careful overlapping window treatment.
- The only explicit 1 day horizon in multimodal text+audio is in [11], where 1 day realized volatility is defined from daily returns; intraday microstructure issues are not addressed.

Thus:

- The existing literature aligns well with “very short multi day” realized volatility (especially 3 day RV), but not with intraday or strictly event window measures.

### 4.2 Implied volatility

- None of the surveyed multimodal ECC papers explicitly uses:
  - Options implied volatility as a target (e.g., near the money short maturity IV).
  - IV changes around calls (IV) or term structure/smile changes.
- All volatility notions in [1,2,3,4,5,6,7,9,10,11,15,16,17,18,19,20] appear to be realized volatility or unspecified “variance” derived from price returns.

For your goal:

- Modeling implied volatility (or IV-adjusted measures like realized minus implied) with multimodal ECC inputs is essentially untouched territory.

## 5. Design implications and opportunities for your research

Given the current state, several clear opportunities emerge:

### 5.1 Targets: move from daily n day to event aligned and IV measures

- Build on existing n day RV targets as a benchmark (1 , 3 , 7 day RV), but extend to:
  - Intraday realized volatility:
    - Call start to close.
    - Pre call vs post call intraday windows.
  - Event window multi day RV:
    - [ 1,+1]or [0,+1] around the call, with precise alignment to call time.
  - Option implied volatility:
    - Short maturity, near the money IV before and after calls.
    - IV as an outcome, potentially capturing revision in volatility expectations due to the call.

No existing multimodal ECC paper has done this, so you would be opening a new axis while still leveraging established multimodal pipelines.

## 5.2 Modeling: combine best ideas from different strands

For text:

- Use LLM based encoders with numeracy aware pretraining:
  - LLM + RAG extraction inspired by ECC Analyzer [6,12].
  - Numeral attachment or numeral aware modules like GNA Vol and NAM [16,17].
  - Dialogue/Q&A structure modeling [18] and sentence selection [19].

For audio:

- Move beyond fixed Praat features:
  - Physics informed or robust paralinguistic models like PIAM [11].
  - Possibly spectrogram based Transformer encoders trained or fine tuned for prosody/emotion.

For fusion:

- Combine:
  - Inter modal attention and stock network awareness from VoTAGE and related works [3,14].
  - Multi source fusion (time series, news, options data) as in RiskLabs [9,10].
  - Affective state representations that map text and audio into joint interpretable spaces [11].

## 5.3 Econometric and evaluation considerations

Borrow from best practices in this literature but tighten them for your more demanding horizons:

- Use:
  - Time based splits and rolling windows as in [5,6,11].
  - Factor only or volatility history baselines, and ablations that:
    - Firm level controls (size, sector).
    - Market/sector volatility, VIX, etc., similar to factor only baselines in [11,21].
- Carefully handle:
  - Microstructure noise in intraday RV (realized kernels, subsampling).
  - Overlapping event windows and serial dependence in volatility.
- Evaluate:
  - On log volatility/variance scale as in [2,3,5,20].
  - Using out of sample  $R^2$  and economic loss functions relevant for trading/risk management.

## 5.4 Fairness and interpretability as first class constraints

- Given the evidence that audio can exacerbate gender bias [5,7,8]:
  - Integrate fairness diagnostics from the start (group level error disparities).
  - Consider adversarial mitigation (AMA LSTM [7]) and multi source fusion (DocFin [8]) to control bias.
- For interpretability:
  - Adopt affective and structural explanations:
    - Affective state features from [11].
    - Attention based visualizations over Q&A segments [18,20].
  - Relate textual and vocal cues to economically interpretable changes in RV or IV.

## 6. Brief practical summary for your project

- If you want “off the shelf” multimodal volatility benchmarks:
  - Start from MDRM [1], HTML [2], and VoTAGE [3] on the MAEC dataset [13].
- If you want state of the art text/audio representation ideas:
  - Use LLM plus RAG and numeracy pretraining from ECC Analyzer [6,12], GNA Vol [16], and NAM [17].
  - Use affective and physics informed audio from The Sound of Risk [11].
- If you care about multi source risk forecasting:

- Look to RiskLabs [9,10] and DocFin [8] as templates for integrating news, time series, and tabular data.
- If you care about fairness and robustness:
  - Incorporate insights and techniques from [5,7,8].

The main gap—and your key opportunity—is to bring these multimodal modeling advances to properly defined, event time aligned intraday or 1 day RV and to options implied volatility around earnings calls, with strict information sets and robust econometric evaluation.

## Categories

### Comparative Overview of Core Papers on Multimodal Earnings Calls for Volatility Prediction

Below I organize and compare the papers along dimensions that are most pertinent to your specific topic: (i) volatility target design and horizon, (ii) use and integration of modalities, (iii) model architectures / learning setups, and (iv) evaluation and econometric discipline. I highlight which are closest to your “very short horizon volatility from text+audio” target, and how they differ.

#### 1. Volatility Targets: Horizon, Definition, and Event Alignment

Key aspects:

- Horizon length and whether they truly focus on “very short horizon”
- Realized vs implied volatility
- Explicit event window design relative to the call

Table 1: Volatility target design across papers

| Ref                                             | Uses ECC text? | Uses ECC audio? | Volatility target type                                              | Horizon(s) (as stated)                                      | Short-horizon focus quality                                         | Implied vol?  | Event-window clarity (relative to call)                                                          |
|-------------------------------------------------|----------------|-----------------|---------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------|
| [1] Qin & Yang 2019                             | Yes            | Yes             | Realized volatility (std dev of returns)                            | $\Delta = 3, 7, 15, 30$ trading days post call              | Strong for 3-day horizon; main emphasis on short-term $\Delta=3$    | No            | Horizon defined in trading days after call; no intraday or precise call time alignment described |
| [2] HTML 2020                                   | Yes            | Yes             | Log realized volatility                                             | 3, 7, 15, 30 days                                           | Strong for 3-day (explicitly “short-term”)                          | No            | Defined as average n day volatility post call; no intraday/event-window details                  |
| [3] VoITAGE 2020                                | Yes            | Yes             | Log volatility $v[0,n]$ and $v_n$ (log abs return) based on returns | n not clearly specified in excerpt; multi-day post call     | Likely similar to 3–30d range; short-horizon but not intraday       | No            | Post call window $[0,n]$ ; no intraday or precise event window description in excerpt            |
| [4] Multimodal Multi-Task Risk Forecasting 2020 | Yes            | Yes             | Volatility + price movement                                         | “Volatility” not precisely defined in excerpt; multi-period | Some short-horizon component but details missing                    | No            | Only “volatility after calls” mentioned, no formal event window in excerpt                       |
| [5] Bias in Multimodal Analysis 2021            | Yes            | Yes             | Average log volatility $v[0,n] = \ln((1/n) (r_i - r)^2)$            | n = 3, 7, 15, 30 days                                       | Strong 3-day focus (explicit short-horizon risk); 3–7d “short-term” | No            | n day window post call; no intraday/call start alignment discussed                               |
| [6,12] ECC Analyzer 2024                        | Yes            | Yes             | “n-day stock volatility”; realized vs implied not specified         | 3, 7, 15, 30 days                                           | Strong short-term emphasis at 3–7d, though definition opaque        | Not specified | Post call n day window implied; event-timing & construction not detailed                         |
|                                                 | Yes            | Yes             | Volatility from dividend adj. daily prices                          |                                                             | Short-term focus suggested                                          | No            | Uses post-call price series; no explicit                                                         |

|                                           |                                  |                                  |                                                              |                                                                                          |                                                                                                |    |                                                                                                   |
|-------------------------------------------|----------------------------------|----------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------|
| [7] AMA LSTM 2024                         |                                  |                                  | (exact formula not given)                                    | Horizon not specified in excerpt                                                         | but not documented                                                                             |    | it window/horizon definition provided                                                             |
| [8] DocFin 2022                           | Yes                              | Yes                              | "Stock volatility" + price movement                          | Horizon / formula not specified                                                          | Includes short-term volatility as a task but unspecified                                       | No | Volatility "after calls"; no event-window description in excerpt                                  |
| [9,10] RiskLabs 2024                      | Yes                              | Yes                              | "Volatility/variance" (multi-task)                           | 3, 7, 15, 30 days                                                                        | Explicit short-/medium-/long-term; strong 3–7d component                                       | No | Motivated by ECC-induced volatility; label construction and precise window not detailed           |
| [11] Sound of Risk 2025                   | Yes                              | Yes                              | Realized volatility (plus CAR)                               | 1, 7, 30-day horizons                                                                    | Includes explicit 1-day realized volatility, closest to "very short-horizon"; also 7 & 30 days | No | Realized volatility over 1, 7, 30 trading days post call; no intraday or [1, +1] design described |
| [13] MAEC dataset 2020                    | Yes                              | Yes                              | "Volatility" in baseline experiments                         | Not specified in excerpt                                                                 | Includes volatility tasks but horizon unspecified                                              | No | Event-window definition not detailed                                                              |
| [14] Risk Forecasting from Acoustics 2020 | Yes (as baseline/feature source) | Yes                              | "Volatility" / "risk"                                        | Not specified                                                                            | Short-term around calls implied but not specified                                              | No | Focus on forecasting volatility but without explicit horizon/measure in excerpt                   |
| [15] NumHTML 2022                         | Yes                              | Yes                              | Volatility as auxiliary task                                 | Horizon / definition not specified                                                       | Volatility helps return prediction; short-term nature likely but undefined                     | No | ECC driven, but event-window not given                                                            |
| [16] GNA Vol 2023                         | Yes                              | (Audio via baselines [1,2])      | "Short-term volatility"                                      | 3-day, 7-day                                                                             | Strong focus on 3–7d short-term                                                                | No | Uses earnings call based labels; detailed construction not in excerpt                             |
| [17] NAM/EC-Num 2021                      | Yes                              | (Audio via baselines [1,2])      | Short-term stock volatility                                  | "Short-term scenarios" (3–7d implied via comparisons)                                    | Focuses on short-term volatility over a few days                                               | No | Earnings-call event-based, but no explicit window definition in excerpt                           |
| [18] Q&A Attention Network 2020           | Yes                              | No                               | Volatility after earnings calls                              | Three "periods" (not named; short/medium/long)                                           | Short-term period present but not quantified                                                   | No | Post ECC volatility, but no explicit horizon or RV formula                                        |
| [19] DeFVP 2024                           | Yes                              | (Time series as second modality) | Volatility prediction                                        | Uses standard earnings call volatility tasks/datasets [1,2,13,15]; horizons likely 3–30d | Short-term components (3–7d)                                                                   | No | Inherits event-windows from base datasets; not detailed here                                      |
| [20] Sparse Attention ECC 2023            | Yes                              | No                               | Average log volatility over n days (3, 15) mapped to classes | n = 3 (short-term), 15 (longer-term)                                                     | Explicit 3-day average log volatility classification                                           | No | Post call 3- and 15-day windows; no intraday alignment                                            |
| [21] PRoFET 2019                          | Yes                              | No                               | Volatility (risk of firms)                                   | Horizon not specified                                                                    | Short-term earnings-call-based volatility but details missing                                  | No | ECC-based volatility; window not described                                                        |

#### Key comparative points:

- Closest to your "very short-horizon" definition:
  - 1-day realized volatility (shortest explicit horizon): [11].

- 3-day realized/log-volatility: [1,2,3,5,6,9,10,16,17,19,20].
- Implied volatility:
  - None of the papers explicitly model options-implied volatility as a target in the provided excerpts; all are realized-volatility or unspecified “volatility/variance” from price data [1,2,3,5,6,7,9,10,11,15,16,17,18,19,20].
- Event-window nuance:
  - Most define horizons in calendar/trading days after the call but do not distinguish pre market, intra day, or post market timing, nor intraday RV vs close-to-close RV [1,2,3,5,6,9,10,11,16,17,18,19,20].
  - [11] is explicit about 1/7/30 trading days but not intraday windows.

For your purposes, [11] is uniquely valuable for a 1-day horizon; [1,2,3,5,6,9,10,16,17,19,20] collectively give a strong picture for 3–7 day post call volatility, but none tackle precise intraday/event-window RV or implied IV.

## 2. Modalities and Feature Engineering: Text vs Audio vs Extras

Key aspects:

- Whether both text and audio are used as predictive inputs (not just transcription)
- Level of granularity and feature types
- Additional modalities (time series, news, tables)

Table 2: Modalities and feature granularity

| Ref                                  | Text processing                                                                                                | Audio processing                                                                                               | Multimodal fusion style                                                                   | Extra modalities                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| [1] Qin & Yang 2019                  | Sentence embeddings (GloVe averaged)                                                                           | 27 Praat features per sentence (pitch, intensity, jitter, HNR, etc.)                                           | Early/intermediate fusion within RNN (Bi LSTM style) sequence model (MDRM)                | Past price-based volatility baseline                            |
| [2] HTML 2020                        | Hierarchical Transformer over sentences                                                                        | 27 Praat acoustic/vocal features aligned at sentence level                                                     | Fusion layer combining text and audio encodings in a multi-task network                   | None beyond price history for baselines                         |
| [3] VoLTAGE 2020                     | FinBERT sentence encoders + inter-utterance attention                                                          | Audio features (not fully listed; likely similar prosodic set) aligned to utterances                           | Cross-modal attention + GCN for inter-stock relationships; semi-supervised multi-task     | Stock correlation graph                                         |
| [4] Multimodal Multi-Task 2020       | “Domain specialized textual features” (likely FinBERT/financial LM)                                            | Audio-attentive alignment, prosodic features                                                                   | Deep multimodal text-audio attention model; multi-task ensemble                           | None beyond price data implied                                  |
| [5] Bias in Multimodal Analysis 2021 | Similar to [2]/[3] setups                                                                                      | Gender-sensitive audio features (prosodic cues) leveraged; multimodal encoders                                 | Multimodal neural models; compare text-only vs audio-only vs full                         | None reported                                                   |
| [6,12] ECC Analyzer 2024             | LLM-based: paragraph-level summaries + RAG extraction of key sentences; textual embeddings                     | Pretrained audio-embedding models; call-level acoustic embeddings                                              | Fusion of LLM-extracted text and audio embeddings; multimodal regression                  | None beyond traditional GARCH/LSTM baselines                    |
| [7] AMA LSTM 2024                    | FinBERT embeddings (as in [5])                                                                                 | Audio features (unspecified, but aligned to text)                                                              | Adversarially trained multimodal LSTM; audio/text components + adversarial perturbations  | None                                                            |
| [8] DocFin 2022                      | Transcript features; prepared vs Q&A segments                                                                  | Explicit prosodic (pitch, intensity) and other audio features                                                  | Multimodal neural fusion of text, audio, and tabular information                          | Semi-structured tabular data from 10 K/10 Q etc.                |
| [9,10] RiskLabs 2024                 | LLM-derived ECC text representations; Earnings Call Encoder                                                    | Acoustic embeddings (vocal cues)                                                                               | Multimodal + multi-source fusion; ablations for Audio+Text vs other combos                | - Pre-call market time series (30 days) - Daily news around ECC |
| [11] Sound of Risk 2025              | LLM-based textual sentiment mapped into 3-D affective state (ASL) features; section- and executive-level stats | Physics-Informed Acoustic Model (PIAM) extracting paralinguistic emotion; ASL features per executive & section | Affective-state fusion: acoustic + text mapped to ASL, then feature engineering + XGBoost | 30-day historical volatility and other financial controls       |
| [13] MAEC 2020                       | Sentence-level linguistic features; word embeddings                                                            | Basic acoustic features + MFCCs; sentence-level alignment                                                      | Baseline multimodal models; details limited in excerpt                                    | None explicitly                                                 |
| [14] Risk from Acoustics 2020        | Uses transcripts for semantic features                                                                         | Speech features; cross inter-modal attention to enforce verbal–vocal coherence                                 | Inter-modal attention + stock-network embeddings                                          | Stock network embeddings (correlation-based)                    |

|                                |                                                                            |                                                           |                                                                |                                             |
|--------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------|
| [15] NumHTML 2022              | Numeric-oriented hierarchical Transformer; special handling of numerals    | Standard Praat acoustic features                          | Multitask fusion of text+audio; volatility as auxiliary        | None beyond price series                    |
| [16] GNA Vol 2023              | Numeral attachment pre-training; LM for EC text                            | Audio appears only via multimodal baselines (MDRM, HTML)  | Main model is text-only; compares to multimodal baselines      | None                                        |
| [17] NAM / ECNum 2021          | Numeral-aware textual model (NAM)                                          | Audio only in multimodal baselines                        | Text-only in main model; multimodal only via comparison        | None                                        |
| [18] Q&A Attention 2020        | Dialogue-form transcript modeling (round-level attention)                  | None                                                      | Text-only                                                      | None                                        |
| [19] DeFVP 2024                | Sentence-level text; differentiable Binary Selector for sentence selection | Time series treated as second modality (price/volatility) | Cross-modal: text selection + time-series volatility predictor | Price/volatility time series                |
| [20] Sparse Attention ECC 2023 | Long-document sparse attention; augmentation with unlabeled sentences      | None                                                      | Text-only; multi-source textual augmentation                   | Financial news as external sentences (text) |
| [21] PRoFET 2019               | Attention over event transcripts                                           | None                                                      | Text + financial features, but no audio                        | Firm-level financial variables              |

Key comparative points:

- “True” multimodal (text + audio) volatility models: [1,2,3,4,5,6,7,8,9,10,11,13,14,15].
- Text-only (but earnings-call based) volatility models: [16,17,18,19,20,21].
- High granularity / sophisticated audio:
  - Physics- and emotion-oriented: [11] (PIAM ' ASL).
  - Sentence-aligned Praat features: [1,2,5,13,15].
  - Spectrogram/embedding based with modern audio encoders: [6,9,10] (pretrained audio models).
- Additional modalities beyond text+audio:
  - Price time-series & news (multi-source): [9,10].
  - Tabular financials: [8].
  - Stock network structure: [3,14].
  - Volatility time series as a separate “modality”: [19].

For your multimodal text+audio focus, [1,2,3,5,6,7,9,10,11,15] are the central group; [8,13,14] add useful variations (document tables, network structure), but with less detail on labels.

### 3. Predictive Setups and Model Architectures

Key aspects:

- Cross-sectional vs time-series (typically pooled panel, but explicit treatment matters)
- Core ML/DL architecture for text+audio
- Single-task vs multi-task setups, auxiliary volatility or returns tasks

Table 3: Predictive setup and architecture

| Ref                                  | Cross-sectional vs time-series                                     | Main architecture                                                       | Task design                                                               | Volatility role             |
|--------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------|
| [1] Qin & Yang 2019                  | Pooled panel over S&P 500 calls (cross-sectional events over time) | RNN-based MDRM (bc-LSTM) over sentence sequences with multimodal inputs | Single-task regression for volatility at horizons $\Delta = 3, 7, 15, 30$ | Primary target              |
| [2] HTML 2020                        | Similar pooled ECC panel                                           | Hierarchical Transformer encoders + multimodal fusion + multi-task head | Joint prediction of average $n$ day log-vol and single-day log-vol        | Primary target (multi-task) |
| [3] VoITAGE 2020                     | Pooled panel + stock graph                                         | FinBERT text encoder + inter-modal attention + GCN (semi-supervised)    | Multi-task: average $v[0, n]$ and single-day $v_n$                        | Primary target              |
| [4] Multimodal Multi-Task 2020       | Pooled panel                                                       | Deep multimodal text-audio attention model                              | Multi-task: volatility + price movement                                   | One of main targets         |
| [5] Bias in Multimodal Analysis 2021 | Pooled panel with gender partitions                                | Multimodal neural nets (similar to [2]/[3])                             | Volatility prediction; bias analysis (error disparities)                  | Primary target              |

|                                |                                                                       |                                                                                                            |                                                                      |                                                |
|--------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------|
| [6,12] ECC Analyzer 2024       | Pooled ECC panel (2017 S&P 500 subset)                                | Pretrained LLM + RAG extractors' embeddings; multimodal fusion regression                                  | Single-task volatility prediction (with multiple horizons)           | Primary target                                 |
| [7] AMA LSTM 2024              | Pooled panel over two datasets                                        | Multimodal LSTM with adversarial training                                                                  | Single-task volatility regression; adversarial robustness & fairness | Primary target                                 |
| [8] DocFin 2022                | Pooled panel                                                          | Fusion network for text, audio, and tabular data                                                           | Multi-task: volatility and price movement                            | Volatility one of main tasks                   |
| [9,10] RiskLabs 2024           | Pooled panel; multi-source inputs include pre-call time series & news | Multi-encoder LLM+DL system (ECC Encoder, News-Market Encoder, Time-Series Encoder) with multimodal fusion | Multi-task: volatility/variance and VaR; multiple horizons           | Volatility is central risk target              |
| [11] Sound of Risk 2025        | Pooled panel; bootstrap sampling for robustness                       | Feature-engineered multimodal ASL features + XG-Boost                                                      | Separate models for CAR and volatility; multiple horizons            | Volatility a primary outcome                   |
| [13] MAEC 2020                 | Pooled panel                                                          | Baseline NN models for multimodal financial risk prediction                                                | Volatility prediction as part of "financial risk" tasks              | Volatility is a key task                       |
| [14] Risk from Acoustics 2020  | Pooled panel with stock network                                       | Cross inter-modal attention + network embeddings                                                           | Volatility prediction                                                | Volatility primary                             |
| [15] NumHTML 2022              | Pooled panel                                                          | Numeric-oriented hierarchical Transformer                                                                  | Multi-task: returns (main) + volatility (auxiliary)                  | Volatility auxiliary, but central for training |
| [16] GNA Vol 2023              | Pooled panel                                                          | Numeral-attachment-pre-trained LM                                                                          | Single-task volatility prediction; ablations vs multimodal baselines | Volatility primary                             |
| [17] NAM / ECNum 2021          | Pooled panel                                                          | Numeral-aware textual model                                                                                | Volatility forecasting; comparisons with prior works [1,2]           | Volatility primary                             |
| [18] Q&A Attention 2020        | Pooled panel                                                          | Multi-Round Q&A Attention Network                                                                          | Volatility prediction across three periods                           | Volatility primary                             |
| [19] DeFVP 2024                | Pooled panel                                                          | Two-module network: Differentiable Binary Selector (text) + Volatility Predictor (time series)             | Single-task volatility forecasting                                   | Volatility primary                             |
| [20] Sparse Attention ECC 2023 | Pooled panel                                                          | Sparse self-attention encoder; counterfactual augmentation                                                 | Classification of 3- and 15-day volatility trend (down/steady/up)    | Volatility main target (categorical)           |
| [21] PProFET 2019              | Pooled panel                                                          | Neural model with attention over transcripts + firm features                                               | Volatility prediction                                                | Volatility primary                             |

#### Key comparative points:

- Almost all papers treat the problem as a pooled panel (events = call instances), effectively cross-sectional per event date with firm-time dependence implicitly learned; explicit firm fixed effects or formal panel econometrics are generally not detailed [1,2,3,5,6,7,9,10,11,15,16,17,18,19,20,21].
- Multi-task learning:
  - Volatility + single-day vs multi-day: [2,3].
  - Volatility + price movement: [4,8].
  - Volatility auxiliary to returns: [15].
  - Volatility + VaR: [9,10].
- Architectures:
  - RNN-based sequence models: [1,3,4,5,7,18].
  - Transformer-based / hierarchical: [2,6,9,10,15,16,19,20].
  - Graph/network enhanced: [3,14].
  - Feature-engineering + gradient boosted trees: [11].
  - LLM-augmented pipelines: [6,9,10].

For an expert comparing modeling approaches, the tension is between:

- Sequence models with sentence-level alignment ([1,2,3]) vs
- Document-level sparse attention ([20]) vs
- Heavy feature engineering with structured pretraining ([15,16,17,19]) vs

- Affective/physics-informed representations ([11]) vs
- LLM-based preprocessing and RAG ([6,9,10]).

#### 4. Evaluation Practices and Econometric Considerations

Key aspects:

- Out-of-sample evaluation and temporal splits
- Benchmarks (e.g., past volatility, GARCH, HAR)
- Controls (pre-call volatility, firm characteristics)
- Bias/fairness analysis

Table 4: Evaluation setup and controls

| Ref                                  | Out-of-sample design                                                                                                       | Key baselines                                                   | Econometric controls / considerations                                                                                                   | Bias / fairness focus                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| [1] Qin & Yang 2019                  | Standard train/val/test split on ECC dataset; details on temporal split not in excerpt                                     | Past volatility, tf-idf + SVR, text-only models                 | Uses past volatility baselines; no explicit firm fixed effects mentioned                                                                | None                                                                       |
| [2] HTML 2020                        | Out-of-sample MSE; multi-task weighting tuned via validation                                                               | Pricing-based and textual baselines, HTML ablations             | No explicit mention of fixed effects; focus on model comparison                                                                         | None                                                                       |
| [3] VoITAGE 2020                     | Comparative & ablation experiments; semi-supervised evaluation                                                             | MDRM, HTML, other multimodal baselines                          | Stock-network GCN to capture inter-stock dependence; no explicit macro controls                                                         | None                                                                       |
| [4] Multimodal Multi-Task 2020       | Quantitative and qualitative analyses vs SOTA                                                                              | Text-only, audio-only, multimodal baselines                     | No explicit mention of volatility history or macro controls                                                                             | None                                                                       |
| [5] Bias in Multimodal Analysis 2021 | Standard train/val/test with 2017 split; evaluation of error disparity                                                     | Text-only and multimodal baselines from prior works [2,3,4]     | Same volatility label as [3]; no extensive macro/factor controls described                                                              | Explicit gender-bias analysis; error disparity by speaker gender           |
| [6,12] ECC Analyzer 2024             | Temporal 80/20 train/test split (time-based)                                                                               | GARCH, LSTM, MDRM, HTML, GPT 4                                  | Comparative performance across 3,7,15,30d; no detailed econometric controls reported                                                    | No explicit bias analysis                                                  |
| [7] AMA LSTM 2024                    | Two datasets; SOTA comparisons with ablations                                                                              | LSTM baseline, SOTA methods ([1,2,3,5,13])                      | Adversarial training for robustness; no detailed macro controls in excerpt                                                              | Focused on reducing gender bias from audio & improving robustness          |
| [8] DocFin 2022                      | Quantitative improvements (5–12%); ablations on modalities                                                                 | Audio-only, text-only, multimodal; prior multimodal models      | Additional tabular financial data used; gender bias reduction measured                                                                  | Explicit bias reduction metrics (gender-related prediction bias)           |
| [9,10] RiskLabs 2024                 | Comparative experiments vs multiple data-source combinations                                                               | Audio+Text, Time Series only, News+Market, etc.                 | Multi-source design implicitly controls for pre-call volatility and news; details on risk-factor controls not given                     | Not emphasized in provided text                                            |
| [11] Sound of Risk 2025              | 50-iteration bootstrap validation; R <sup>2</sup> measures; ablations (Factors-only, Acoustic-only, Text-only, Multimodal) | Factor-only (historical volatility & controls), unimodal models | Uses firm-level financial factors including 30-day historical volatility; interpretable decomposition (CFO vs CEO, presentation vs Q&A) | Not explicitly bias-focused, but analyzes roles of exec roles and sections |
| [13] MAEC 2020                       | Baseline MSE across years and firm-size groups                                                                             | Price-based methods, textual baselines                          | Notes performance variation by firm size; not much detail in excerpt                                                                    | None                                                                       |
| [14] Risk from Acoustics 2020        | Ablation and case study                                                                                                    | SOTA methods from text and speech                               | Incorporates stock-network correlations; no explicit macro controls                                                                     | None                                                                       |
| [15] NumHTML 2022                    | MSE, trading simulation; Pareto optimality for multi-task                                                                  | HTML, other multimodal baselines                                | Focus on numeric understanding; no explicit risk-factor controls discussed                                                              | None                                                                       |
| [16] GNA Vol 2023                    | MSE on 3- and 7-day tasks; ablations on pretraining                                                                        | MDRM, HTML, FinBERT, NAM                                        | Compares discrete vs continuous labeling strategies                                                                                     | None                                                                       |
| [17] NAM / ECTum 2021                |                                                                                                                            |                                                                 |                                                                                                                                         | None                                                                       |

|                                |                                                                                         |                                            |                                                                                                        |      |
|--------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------|------|
|                                | Volatility forecasting performance vs [1,2]                                             | MDRM, HTML, other text models              | Focus on numerals; no econometric controls mentioned                                                   |      |
| [18] Q&A Attention 2020        | Volatility prediction across three periods; outperforms previous SOTA                   | PRoFET [21], MDRM [1], HTML [2]            | Emphasizes dialogue structure; no explicit risk-factor controls                                        | None |
| [19] DeFVP 2024                | Extensive experiments on three datasets                                                 | Multimodal ECC baselines ([1,2,3,4,13,15]) | Investigates impact of text selection on time-series forecasting; treats price series as core modality | None |
| [20] Sparse Attention ECC 2023 | Performance on 3-class volatility direction for 3 & 15d; interpretability via attention | Multimodal baselines including [1,2,5]     | Uses textual augmentation but not financial controls                                                   | None |
| [21] PRoFET 2019               | Out-of-sample comparisons vs financial-feature baselines                                | Financial-only models                      | Joint use of transcripts + financial features; provides token-level attention interpretability         | None |

Key comparative points:

- Stronger temporal discipline:
  - Explicit time-based train/test split: [5] (2017 split), [6] (80/20 temporal split), likely [1,2,3] though not detailed in excerpt.
  - Bootstrap-based robust validation: [11].
- Econometric controls:
  - Explicit financial controls (e.g., historical volatility): [11,21]; [11] stands out for structured factor-only vs multimodal comparisons.
  - Use of pre-call time series and news as inputs: [9,10] (less a “control” and more an additional predictive modality).
- Bias/fairness:
  - Direct, in-depth gender bias analysis: [5,7,8].
  - Bias mitigation via adversarial training: [7].
  - Multimodal designs shown to both improve accuracy and mitigate or exacerbate bias depending on architecture: [5,7,8].

## 5. Synthesis: Top Findings and Comparative Takeaways for Your Topic

Bringing the comparisons together, the most relevant observations for your specific research goal:

1. Core multimodal (text + audio) volatility prediction benchmark line:

- [1] Qin & Yang 2019: First widely cited multimodal text+audio earnings call model for short-term realized volatility (3–30d), with sentence-level alignment and RNN fusion, demonstrating significant gains over text-only and price-only baselines, especially at 3 days.
- [2] HTML 2020: Hierarchical Transformer extension with multi-task (average vs single-day) volatility, again with multimodal gains most pronounced at short horizons (3d).
- [3] VoLTAGE 2020: Adds cross-modal attention and GCN-based modeling of stock interdependencies, improving on [1,2] for similar horizons.
- [5] Bias in Multimodal 2021: Shows that these multimodal gains can be uneven across subgroups (gender), highlighting fairness considerations in multimodal volatility prediction.

1. Modern LLM and multimodal fusion developments:

- [6,12] ECC Analyzer 2024 and [9,10] RiskLabs 2024: Use large LMs and RAG for structured textual representation, fused with pretrained audio embeddings, and show further improvements over [1,2] especially for 3- and 7-day horizons. They also integrate additional data sources (time series, news) in [9,10], making them closest to a comprehensive risk-forecasting system while still ECC-centric.
- [11] Sound of Risk 2025: Introduces a physics-informed acoustic pipeline and an affective-state representation (ASL) that yields very strong explanatory power for 7–30d volatility and includes explicit 1 day realized volatility, though it focuses on feature engineering + XGBoost rather than end-to-end deep sequence models.

1. Specialized angles important for your design:

- Numeral understanding for volatility: [15,16,17] show that better treatment of numerals in ECC text can materially improve short-term volatility prediction, often exceeding multimodal baselines on text-only setups. Their volatility tasks use the same 3–7d horizons as [1,2].
- Dialogue structure: [18] indicates that modeling the Q&A structure provides incremental predictive value for volatility beyond bag-of-words/sequence models.

- Text selection as cross-modal factor: [19] shows that intelligent sentence selection (via Differentiable Binary Selector) can significantly enhance volatility forecasting when combining textual and time-series modalities.
1. Fairness and robustness:
- [5,7,8] collectively demonstrate that:
    - Audio is a strong contributor to predictive performance but also a source of gender bias.
    - Multimodal fusion can either amplify or mitigate bias depending on training and architecture.
    - Adversarial schemes ([7]) and adding alternative modalities (tabular data; [8]) can reduce bias without sacrificing accuracy.
1. Gaps relative to your ideal specification:
- Precise very short-horizon event windows (intraday, call-start–close, [ 1,+1]) are not explicitly modeled in these works; horizons are almost always in trading days (1,3,7,15,30) with close-to-close returns.
  - Options-implied volatility is essentially absent as a target; all work is on realized volatility from prices or an unspecified “volatility/variance” derived from price/return series.
  - Explicit panel designs with firm/time fixed effects or hierarchical modeling are uncommon; most studies rely on pooled DL models.

In short, the main “top line” for your topic is the evolution from:

- Early multimodal RNN-based models for 3–30d realized volatility from ECC text+audio [1,2,3], toward
- LLM-augmented, physics-informed, multi-source risk systems with improved performance, interpretability, and some attention to fairness [5,6,7,8,9,10,11].

For an expert researcher, the tables above should give a clear basis to:

- Select appropriate benchmark models and datasets ([1,2,3,13]),
- Identify which horizons and target constructions are closest to your intended very short-event-window focus ([1,2,3,5,6,9,10,11,16,17,20]),
- And pinpoint where the literature is thinnest (intraday/event-window RV, implied volatility, explicit panel structures), thereby defining where your contributions can be most novel.

## Timeline

### Chronological evolution of the literature

#### 1. Pre 2019 foundations: text only earnings call volatility and event transcripts

Before the multimodal wave, most work used only text from earnings calls and related events to forecast volatility.

- **PRoFET (2019)** is an important early neural text only benchmark: it uses earnings call transcripts plus financial features to predict volatility with an attention based architecture over language representations [21].
  - It establishes (i) short horizon volatility prediction as a core task for earnings call NLP, and (ii) the usefulness of neural language models plus financial controls, but it does not use audio.

Around the same time, there is also text only work emphasizing dialogue/Q&A structure [18] and volatility prediction from transcripts (not audio). These create the baseline that later multimodal models compare against.

Key takeaway for this stage:

- Earnings call text is recognized as useful for volatility forecasting, but acoustic/voice information is not yet systematically integrated.

#### 2. 2019: First true multimodal text+audio volatility models

2019 marks the crucial multimodal “birth” of the field.

- **Qin & Yang – “What You Say and How You Say It Matters” (ACL 2019) [1]:**
  - First widely cited multimodal model directly targeting stock volatility using both transcripts and audio.
  - Constructs a sentence aligned dataset of 576 S&P 500 calls with:
    - 300d averaged GloVe embeddings for text.
    - 27 acoustic features (pitch, intensity, jitter, HNR, etc.) from Praat for audio.
  - Targets: short horizon post call volatility (3, 7, 15, 30 days), with strongest gains at 3 days [1].

Architecture: RNN (bc LSTM style) sequence model over sentences and a multimodal deep regression model (MDRM) fusing text+audio.

- Provides a public dataset and baseline that almost all subsequent multimodal papers build on or cite [2,3,4,6,7,8,9,10,12,13,14,15,16,17,18,19].

Impact:

- Establishes:
  - Short term realized volatility (3 day log vol/sd) as the canonical label.
  - Sentence level text–audio alignment as a practical unit.
  - Prosodic features as economically meaningful (“what you say and how you say it”).

This paper is the first major milestone for your specific topic: multimodal ML/DL volatility prediction from earnings calls.

### 3. 2020: Rapid expansion – Transformers, multi task learning, GCNs, and larger datasets

2020 sees an explosion of follow up work along several directions:

#### 1. Hierarchical Transformers and multi task volatility (HTML)

- **HTML (The Web Conference 2020) [2]:**
  - Moves from RNNs to **hierarchical Transformers** for text.
  - Still uses Praat acoustic features and multimodal fusion, but adds:
    - Multi task learning: jointly predicts average n day volatility and single day volatility for  $n = 3, 7, 15, 30$ .
    - Explicit log volatility targets.
  - Reports 17–49% improvement over baselines, with audio especially helpful for 3 day volatility [2].

Conceptual shifts:

- Transformer based hierarchical encoders become the standard for long earnings call transcripts.
- Multi task setups (short vs long horizon, averaged vs single day) are normalized.

#### 1. Graph based inter stock relations and attention based multimodal fusion

- **VoITAGE (EMNLP 2020) [3]:**
  - Uses FinBERT for text, learned vocal features, and **inter modal attention** for verbal–vocal coherence.
  - Adds a **Graph Convolution Network (GCN)** to capture stock interdependencies.
  - Targets: log volatility over n day windows post call (average  $v[0,n]$  and single day  $v_n$ ) [3].
- **Risk Forecasting from Earnings Calls Acoustics and Network Correlations (Interspeech 2020) [14]:**
  - Similar spirit: cross inter modal attention for verbal–vocal coherence + stock network embeddings.
  - Focuses on risk/volatility prediction, deepens the idea that volatility is shaped both by firm specific voice/text and cross firm network effects.

#### 1. Multimodal multi task “financial risk” and multi objective setups

- **Multimodal Multi Task Financial Risk Forecasting (ACM MM 2020)[4]:**
  - Jointly predicts volatility and price movement using a deep text audio attention model.
  - Reinforces volatility + returns as complementary risk dimensions in multi task settings.

#### 1. Larger, aligned multimodal datasets

- **MAEC dataset (CIKM 2020) [13]:**
  - Major data milestone: 3,443 call instances from 1,213 S&P 1500 firms (2015–2018) with sentence level text–audio alignment.
  - Provides linguistic features, acoustic features, MFCCs, and baseline volatility prediction experiments [13].
  - Enables scaling beyond the ~500 call ACL19 dataset.

Impact of 2020 wave:

- The field standardizes around:
  - 3–30 day realized post call volatility as the main target.
  - Sentence level text audio fusion using Praat like prosodic features.
  - Advanced sequence encoders (Transformers) and multi task objectives.

- Incorporating network structure among stocks via GCNs.

This is the second major milestone: multimodal volatility prediction becomes a full-fledged research niche with multiple architectures and datasets.

#### 4. 2020–2021: Specialization – Q&A structure, numerals, and fairness/bias

After the initial multimodal architectures, research diversifies to refine specific aspects.

##### 1. Dialogue structure and Q&A emphasis (still text only but volatility focused)

- **Multi Round Q&A Attention Network (IJCAI 2020)** [18]:
  - Focuses on Q&A dialogue structure in transcripts for volatility prediction over multiple periods.
  - Shows that modeling turn taking and dialogue rounds improves volatility forecasts relative to word/document level models [18].
  - Still text only, but influences later multimodal work's attention to Q&A vs prepared remarks.

##### 1. Numeral understanding for volatility

- **ECNum + NAM (CIKM 2021)** [17]:
  - Introduces ECNum, a dataset for numeral types (monetary, percentages, dates, etc.) in earnings call transcripts.
  - Proposes a Numeral Aware Model (NAM) and shows better volatility forecasting in short term scenarios [17].
  - Emphasizes that fine grained numerical semantics in text can improve volatility prediction, independent of audio.

##### 1. Bias and fairness in multimodal volatility models

- **An Empirical Investigation of Bias in the Multimodal Analysis of Financial Earnings Calls (NAACL 2021)** [5]:
  - Uses multimodal volatility models similar to VoTAGE/HTML on S&P 500 calls.
  - Explicitly studies gender bias:
    - Defines volatility labels  $v[0, \Delta]$  with  $\Delta \in \{3, 7, 15, 30\}$ .
    - Shows that multimodal models can outperform text only but also amplify gender related error disparities, partly due to gender sensitive acoustic features and underrepresentation of female executives [5].
  - Releases code and data splits by gender, catalyzing the fairness sub line.

Impact:

- The field starts to move from “can we predict volatility?” to:
  - “How do specific linguistic structures (Q&A, numerals) and acoustic cues contribute?”
  - “Are these models fair across demographic groups?”

#### 5. 2022–2023: Numeral focused pretraining, tabular integration, and text only refinements

##### 1. More advanced numeral understanding tied to volatility

- **GNA Vol and ECNum extensions (IJCNLP 2023)** [16]:
  - Proposes a General Numeral Attachment pretraining (GNA Vol):
    - Enhances language models' ability to attach numerals to the correct entities/contexts in earnings calls.
    - Shows improved 3 and 7 day volatility forecasting vs MDRM, HTML, FinBERT, NAM [16].
  - Conceptually: moves from ad hoc numeral features to task driven pretraining with volatility as downstream objective.

##### 1. Multimodal + semi structured/tabular data

- **DocFin (EMNLP 2022)** [8]:
  - Combines earnings call audio + text with semi structured document tables (10 K/10 Q, etc.).
  - Demonstrates 5–12% improvement in volatility and price movement prediction and reduces gender related bias by over 30% [8].
  - Represents a move toward “realistic” multi source financial prediction systems.
- **NumHTML (AAAI 2022)** [15]:
  - Extends HTML with numeric oriented hierarchical Transformers and multi task learning; volatility becomes an auxiliary task to return prediction.

- Uses text + audio from MAEC style data [15].
- Shows how numeracy in language models boosts both price and volatility forecasts.

### 1. Text only architectures with rich augmentation

- **Sparse Attention + Counterfactual Augmentation (AIFinance 2023)** [20]:
  - Text only, but directly predicts future volatility classes (down/steady/up) from transcripts, targeting average log volatility over 3 and 15 days [20].
  - Introduces multi source counterfactual augmentation using external unlabeled sentences and sparse attention for long transcripts.
  - While audio is absent, this work is a strong text only volatility baseline that multimodal models must improve upon.

#### Impact:

- The key theme becomes “feature quality” and “multi source integration”:
  - Numeral understanding and semi structured financial tables.
  - More sophisticated pretraining and augmentation.
- Volatility remains short horizon (3–30 days realized) and call event based, consistent with your target topic, though still not intraday/IV focused.

## 6. 2020–2024: Dataset and methodological refinements; robustness and denoising

- **MAEC’s influence** continues: many models (NumHTML [15], DocFin [8], AMA LSTM [7]) build on MAEC or similar S&P 500/1500 aligned ECC datasets [13].
- **DeFVP (ICME 2024)** [19]:
  - Formally treats time series and text as two modalities and investigates how sentence level textual selection (via a Differentiable Binary Selector) affects volatility forecasting.
  - Demonstrates that intelligently “denoised” textual input significantly helps time series volatility models [19].
- These works show a trend toward:
  - Explicitly modeling the interaction between textual “noise” and financial time series signals.
  - Moving beyond simple “feed all sentences” to more selective and interpretable use of text.

## 7. 2024–2025: LLM centric multimodal frameworks and physics informed acoustics

More recent work marks a qualitative shift: large language models and specialized acoustic models enter the pipeline, and the focus broadens from pure accuracy to interpretability and robustness.

### 1. LLM based ECC analyzers and multi source risk frameworks

- **ECC Analyzer (AIFinance 2024) and its arXiv version** [6,12]:
  - Uses LLMs to:
    - Summarize earnings call paragraphs.
    - Drive RAG with a finance “Question Bank” to extract key sentences.
  - Extracts text and audio embeddings via pretrained models and fuses them to predict n day volatility, with strong gains at 3–7 days over classical GARCH, LSTM, and previous multimodal baselines like MDRM and HTML [6,12].
  - Conceptual shift: LLMs become upstream feature engineers; downstream volatility models remain relatively standard regressors.
- **RiskLabs (2024, arXiv + journal version)** [9,10]:
  - Multi source multimodal pipeline:
    - Earnings call audio + transcripts.
    - 30day pre call price time series.
    - Firm centered news around ECC dates.
  - Uses LLM derived textual embeddings, acoustic encoders, and a fusion network to perform multi task forecasting of volatility/variance and VaR over 3, 7, 15, 30 days [9,10].
  - Demonstrates:
    - LLMs alone are not sufficient as end to end predictors but are powerful for representation learning.

- Multi source fusion can meaningfully improve volatility and risk metrics.

## 1. Robustness, fairness, and adversarial training in multimodal models

- **AMA LSTM (2024)**[7]:
  - Adversarially trained multimodal LSTM using FinBERT text embeddings and audio features.
  - Targets stock volatility (short horizon, realized from dividend adjusted prices) and explicitly injects adversarial perturbations to simulate market stochasticity and reduce gender bias [7].
  - Reports that:
    - Audio carries more gender bias than text.
    - Multimodal fusion and adversarial training improve robustness and fairness.

This connects to the earlier NAACL 2021 bias work [5] but now with a more explicit adversarial mitigation framework.

## 1. Physics informed acoustic modeling and affective state representations

- **The Sound of Risk (2025)** [11]:
  - Large sample (H1,795 calls, 1,800 hours) with a focus on acoustic emotion and interpretability.
  - Uses a Physics Informed Acoustic Model (PIAM) to extract robust vocal emotional features and map them, along with LLM based text sentiment, into a 3 D Affective State Label (ASL: Tension, Stability, Arousal) [11].
  - Engineers H150 features per executive/call section, capturing:
    - Moments and changes of ASL.
    - Differences between prepared remarks and Q&A (presentation'Q&A shifts).
  - Predicts realized volatility over 1-, 7-, 30 day horizons and cumulative abnormal returns using boosted trees (XGBoost).
  - Main findings:
    - Multimodal affective features explain up to 43.8% of out of sample variance in 30 day realized volatility [11].
    - Predictive power is concentrated in affective shifts during the transition to Q&A.
  - Major conceptual shift:
    - From generic prosodic features to a structured affective state representation with a physics informed speech model.
    - Emphasis on interpretability (what aspects of vocal emotion drive volatility) rather than only predictive accuracy.

Impact of 2024–2025 wave:

- The field moves to:
  - LLM based preprocessing and RAG to “understand” calls.
  - Physics informed and domain specific speech models for more robust acoustic features.
  - Multi source (calls + news + time series + tabular) and multi task (volatility, VaR, returns) risk forecasting.

Still, horizons remain predominantly 3–30 days realized volatility; intraday or strictly event window RV and option implied volatility remain largely unexplored in this multimodal ECC context.

## Thematic trends and patterns over time

### 1. Targets and horizons: from generic “risk” to standardized 3–30 day realized volatility

- Early work used “risk” somewhat loosely; PRoFET [21] and early transcript work adopted volatility as a natural metric but with varied horizons.
- Qin & Yang [1], HTML [2], VoLTAGE [3], and subsequent works converged on:
  - Realized volatility from returns over fixed n day windows (n=3,7,15,30) after the call.
  - Sometimes log volatility or average log volatility  $v[0,n]$  as the label.
- Later work (ECC Analyzer [6,12], RiskLabs [9,10], GNA Vol [16]) mostly inherits these horizons.

Notably:

- There is almost no explicit use of:
  - Intraday realized volatility aligned precisely to call start/finish.
  - Option implied volatility surfaces tied to calls.

That leaves an open frontier for the kind of very short horizon (intraday, [ 1,+1], next day RV) and IV based work you're interested in.

## 2. Modeling approaches: RNNs ' hierarchical Transformers ' LLMs and structured fusion

- 2019–2020:
  - RNNs/LSTMs with averaged GloVe embeddings [1].
  - Hierarchical Transformers (HTML [2]) and FinBERT encoders (VoTAGE [3]).
- 2020–2022:
  - GCNs for stock networks [3,14].
  - Multi task and Pareto multi objective training (HTML [2], NumHTML [15]).
- 2022–2025:
  - LLMs (FinBERT, GPT class models) for representation learning and RAG [6,9,10,11,12].
  - Task specific pretraining (GNA Vol [16]) and numeral aware models [17].
  - Physics informed acoustic models and affective state mapping [11].
  - Denoising/selection mechanisms recognizing text as a noisy modality [19].

Trend:

- The text pipeline mirrors the general NLP evolution: RNNs ' Transformers ' LLMs and specialized pretraining.
- The audio pipeline moves more slowly:
  - From handcrafted Praat features [1,2] to richer paralinguistic features and network aware models [3,8,11,14], with PIAM [11] being the most sophisticated so far.

## 3. Multimodal fusion: early concatenation to attention and joint affective spaces

- Early multimodal models:
  - Early fusion via concatenation of text and audio embeddings or features at sentence/call level [1,2].
- 2020:
  - Inter modal attention and verbal–vocal coherence mechanisms (VoTAGE [3], Interspeech 2020 [14]) become standard.
  - GCNs to jointly propagate information over firm networks.
- 2022–2025:
  - Joint modeling with additional modalities (tabular data [8], time series, news [9,10]).
  - Affective state representations where text and audio jointly define ASL coordinates [11].
  - LLM guided RAG to select and emphasize important textual segments [6,12,19].

Trend:

- Fusion moves from simple concatenation to:
  - Cross attention and coherence modeling.
  - Hierarchical, multi stage fusion with structured semantic and affective spaces.
  - Integration of more modalities (news, tables, time series).

## 4. From “black box” prediction to interpretability and fairness

- Early works focus purely on predictive accuracy.
- From 2021 onward:
  - Explicit fairness and bias analyses:
    - Gender bias in multimodal volatility models [5].
    - Bias mitigation via adversarial training and multimodal fusion [7].
    - Bias reduction via adding tabular data [8].
  - Interpretability:
    - Token level attention visualization (PRoFET [21]).
    - Q&A specific attention [18].
    - Affective state explanations (which vocal and textual affective changes drive volatility) [11].

Trend:

- The community increasingly treats volatility forecasts as risk tools that must be:
  - Fair across demographic groups.
  - Interpretable enough for financial and regulatory scrutiny.

## Key research clusters and sustained contributors

Several author groups form clear “threads” in this literature.

### 1. Qin–Yang–Dong / MAEC–HTML–NumHTML cluster

Core works:

- Qin & Yang ACL2019 (MDRM) [1].
- HTML (The Web Conference 2020) [2].
- MAEC dataset (CIKM 2020) [13].
- NumHTML (AAAI 2022) [15].

Contributions:

- Provided the foundational multimodal dataset and first widely used multimodal volatility model [1].
- Pioneered hierarchical Transformer architectures and multi task volatility objectives [2].
- Scaled to a large, public, sentence aligned ECC dataset (MAEC) [13].
- Advanced numeracy and multi task modeling with NumHTML [15].

Implication:

- This group has effectively set the “standard tasks and data” for multimodal volatility prediction. Future work on very short horizon RV/IV could naturally extend MAEC or follow their evaluation protocols.

### 2. Sawhney–Shah / VoLTAGE–multimodal risk–bias cluster

Core works:

- VoLTAGE (EMNLP 2020) [3].
- Multimodal Multi Task Financial Risk Forecasting (ACM MM 2020) [4].
- Risk forecasting with acoustics and network correlations (Interspeech 2020) [14].
- Bias in multimodal analysis (NAACL 2021) [5].

Contributions:

- Introduced inter modal attention, CEO focused vocal cues, and stock network GCNs [3,14].
- Pushed multi task risk forecasting including volatility and price movement [4].
- Highlighted and quantified gender bias and demographic concerns in multimodal volatility models [5].

Implication:

- This line has shaped the notion of “rich multimodality” (network + audio + text) and fairness in volatility prediction. It suggests future work will need to address both network structure and fairness when moving to finer horizons or options markets.

### 3. Chen–Hsin-Hsi / numerals and pretraining (ECCNum, GNA Vol)

Core works:

- ECCNum + NAM for volatility (CIKM 2021) [17].
- GNA Vol (IJCNLP 2023) [16].

Contributions:

- Recognized numeral understanding as a bottleneck in ECC based volatility forecasting.
- Produced datasets and models that specialized in numerals and demonstrated that numeral specific pretraining improves short term volatility prediction [16,17].

Implication:

- For intraday or IV based volatility, numerical details (option deltas, vol quotes, EPS surprises, guidance) are crucial; this group’s methods could be directly leveraged in that context.

### 4. Cao–Ndiaye / LLM based ECC Analyzer and RiskLabs

Core works:

- ECC Analyzer (AIFinance 2024 + arXiv) [6,12].

- RiskLabs (2024, arXiv + journal) [9,10].

#### Contributions:

- Positioned large language models as central to extracting trading signals from ECCs for volatility prediction.
- Integrated multi source data (calls, news, pre call time series) and multi task risk metrics (volatility, variance, VaR) [9,10].
- Benchmarked against earlier multimodal models (MDRM, HTML) and GARCH/LSTM baselines, showing the incremental value of LLMs and additional data sources [6].

#### Implication:

- This cluster points toward a future in which volatility forecasts are generated by complex, multi source, LLM augmented systems rather than standalone ECC models.

## 5. Robustness, fairness, and advanced audio – AMA LSTM, DocFin, Sound of Risk

#### Core works:

- AMA LSTM (2024) [7].
- DocFin (EMNLP 2022) [8].
- The Sound of Risk (2025) [11].

#### Contributions:

- Demonstrated adversarial training and multimodal fusion as tools to mitigate gender bias and increase robustness [7].
- Showed that adding semi structured/tabular data to audio+text improves volatility prediction and reduces bias [8].
- Introduced physics informed acoustic modeling and affective state representations to explain volatility [11].

#### Implication:

- This group of works suggests a clear future direction: more robust, interpretable, and fairness aware multimodal volatility models, potentially applicable to regulatory or institutional settings.

## Gaps and implications for future very short horizon RV/IV research

From the timeline, several consistent patterns—and gaps—emerge:

- Volatility horizons:
  - Almost all multimodal ECC work uses 3–30 day realized volatility from daily returns [1,2,3,4,5,6,8,9,10,11,15,16,17,18,20].
  - Explicit intraday RV, call start–close windows, or [ 1,+1] event windows are virtually absent.
  - None clearly model option implied volatility around ECCs (levels or changes) as a primary target.
- Label alignment:
  - Most papers align horizons relative to “call date” rather than precise call time (pre /post market or intraday).
  - This is acceptable for 3–30 day horizons but would be problematic for the ultra short horizons you care about.
- Econometric sophistication:
  - Past volatility controls, log variance transforms, and multi task setups are common.
  - Detailed microstructure corrections, realized kernels, or jump decompositions for HF RV are not used.
- Multimodal sophistication:
  - Text: already at LLM+RAG and task specific pretraining (GNA Vol) [6,9,10,12,16].
  - Audio: moving toward physics informed and affective modeling [11], but there is still room for:
    - End to end spectrogram Transformers.
    - Precise alignment with utterances and event timing relevant for intraday volatility.

Given this history, the next major milestone for the field that would align with your goal likely involves:

- Constructing high frequency realized volatility and/or option implied volatility labels tightly aligned to call timing.
- Applying LLM enhanced, numeracy aware text encoders and advanced acoustic models (like PIAM or spectrogram Transformers) in a multimodal fusion pipeline.
- Embedding these into:
  - Panel/cross sectional settings (e.g., “next day RV for all firms with calls today”) and
  - Firm time series settings (across multiple quarters per firm).
- Explicitly addressing fairness and interpretability, building on the bias and affective state literature [5,7,8,11].

The current literature provides strong methodological building blocks and datasets, but leaves the very short horizon and implied volatility dimension largely open.

## Foundational Work

### Which papers form the foundational references on this topic?

The below table shows the resources that are most often cited by the relevant papers on this topic. This is measured by the **reference rate**, which is the fraction of relevant papers that cite a resource. Use this table to determine the most important core papers to be familiar with if you want to deeply understand this topic. Some of these core papers may not be directly relevant to the topic, but provide important context.

| Ref.  | Reference Rate | Title                                                                                                            | Cited By These Relevant Papers             |
|-------|----------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| [2]   | 0.89           | HTML: Hierarchical Transformer-based Multi-task Learning for Volatility Prediction                               | [3, 4, 5, 6, 7, 8, 10, 12, 13, 14, 15, 16] |
| [1]   | 0.73           | What You Say and How You Say It Matters: Predicting Stock Volatility Using Verbal and Vocal Cues                 | [2, 3, 4, 6, 7, 8, 9, 10, 12, 13, 15, 16]  |
| [119] | 0.58           | The sound of confidence and doubt                                                                                | [1, 2, 3, 4, 5, 13, 14]                    |
| [112] | 0.57           | Predicting Risk from Financial Reports with Regression                                                           | [1, 2, 3, 4, 5, 7, 8, 12, 15]              |
| [105] | 0.50           | Volatility Prediction using Financial Disclosures Sentiments with Word Embedding-based IR Models                 | [1, 2, 6, 7, 12, 13, 15]                   |
| [59]  | 0.48           | A Semiparametric Gaussian Copula Regression Model for Predicting Financial Risks from Earnings Calls             | [1, 2, 3, 5, 6, 12, 15]                    |
| [127] | 0.48           | Deep Learning for Event-Driven Stock Prediction                                                                  | [1, 2, 3, 6, 13, 15]                       |
| [122] | 0.48           | Volatility forecast using hybrid Neural Network models                                                           | [2, 3, 6, 7, 14, 15]                       |
| [66]  | 0.42           | Modeling Financial Analysts' Decision Making via the Pragmatics and Semantics of Earnings Calls                  | [3, 4, 5, 13, 15]                          |
| [13]  | 0.36           | MAEC: A Multimodal Aligned Earnings Conference Call Dataset for Financial Risk Prediction                        | [6, 7, 8, 15]                              |
| [3]   | 0.34           | VolTAGE: Volatility Forecasting via Text-Audio Fusion with Graph Convolution Networks for Earnings Calls         | [5, 7, 8]                                  |
| [131] | 0.33           | Twitter mood predicts the stock market                                                                           | [2, 3, 6, 13]                              |
| [141] | 0.32           | Hierarchical Attention Networks for Document Classification                                                      | [2, 4, 14, 15]                             |
| [142] | 0.32           | Detecting Deceptive Discussions in Conference Calls                                                              | [2, 4, 13, 15]                             |
| [143] | 0.32           | Listening to Chaotic Whispers: A Deep Learning Framework for News-oriented Stock Trend Prediction                | [3, 4, 13, 15]                             |
| [30]  | 0.32           | Stock Volatility Prediction Based on Self-attention Networks with Social Information                             | [2, 3, 14, 15]                             |
| [109] | 0.27           | Applications of deep learning in stock market prediction: recent progress                                        | [3, 4, 14]                                 |
| [58]  | 0.25           | Financial Keyword Expansion via Continuous Word Vector Representations                                           | [1, 2, 4]                                  |
| [144] | 0.25           | The Power of Voice: Managerial Affective States and Future Firm Performance                                      | [1, 2, 4]                                  |
| [45]  | 0.25           | Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models | [2, 4, 6, 12]                              |

## Adjacent Work

### Which papers cite the same foundational papers as relevant papers?

Use this table to discover related papers on adjacent topics, to gain a broader understanding of the field and help generate ideas for useful new research directions.

| Ref.  | Adjacency score | Title                                                                                                                                  | References These Foundational Papers |
|-------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| [46]  | 2.98            | Multimodal Multi-Speaker Merger & Acquisition Financial Modeling: A New Task, Dataset, and Neural Baselines                            | [1, 2, 3, 4, 5, 105, 127]            |
| [74]  | 2.72            | Measuring Consistency in Text-based Financial Forecasting Models                                                                       | [1, 2, 4, 5, 13, 105, 127]           |
| [31]  | 2.55            | KeFVP: Knowledge-enhanced Financial Volatility Prediction                                                                              | [1, 2, 3, 4, 13, 127]                |
| [151] | 2.26            | Language Modeling for the Future of Finance: A Quantitative Survey into Metrics, Tasks, and Data Opportunities                         | [1, 2, 3, 7, 13]                     |
| [38]  | 2.23            | Same Company, Same Signal: The Role of Identity in Earnings Call Transcripts                                                           | [1, 2, 3, 4, 13, 18]                 |
| [32]  | 2.19            | Predicting Corporate Risk by Jointly Modeling Company Networks and Dialogues in Earnings Conference Calls                              | [1, 2, 3, 13, 18, 112, 124]          |
| [19]  | 2.10            | How Does Textual Information Selection Influence Time Series Forecasting? A Cross-modal Perspective on Financial Volatility Prediction | [1, 2, 3, 4, 13]                     |
| [43]  | 2.09            | ECHO-GL: Earnings Calls-Driven Heterogeneous Graph Learning for Stock Movement Prediction                                              | [1, 2, 3, 4, 13]                     |
| [152] | 1.94            | Extracting key insights from earnings call transcript via information-theoretic contrastive learning                                   | [1, 2, 3, 13, 18]                    |
| [153] | 1.58            | Predicting bankruptcy of firms using earnings call data and transfer learning                                                          | [1, 2, 13, 59]                       |
| [37]  | 1.52            | Speech emotion recognition and text sentiment analysis for financial distress prediction                                               | [2, 3, 4, 5]                         |
| [33]  | 1.38            | Stock Price Volatility Prediction: A Case Study with AutoML                                                                            | [1, 13, 105, 112, 131]               |
| [54]  | 1.32            | Modal-adaptive Knowledge-enhanced Graph-based Financial Prediction from Monetary Policy Conference Calls with LLM                      | [1, 2, 4]                            |
| [61]  | 1.29            | MONOPOLY: Financial Prediction from MONetary POLicY Conference Videos Using Multimodal Cues                                            | [1, 2, 4]                            |
| [135] | 1.28            | Unlocking Financial Insights: An advanced Multimodal Summarization with Multimodal Output Framework for Financial Advisory Videos      | [2, 4, 13]                           |
| [39]  | 1.26            | Causal Inference from Text: Unveiling Interactions between Variables                                                                   | [1, 3, 5, 59]                        |
| [125] | 1.23            | Incorporating Stock Market Signals for Twitter Stance Detection                                                                        | [1, 4, 5, 105]                       |
| [28]  | 1.21            | Multi-Domain Transformer-Based Counterfactual Augmentation for Earnings Call Analysis                                                  | [2, 3, 5]                            |
| [72]  | 1.21            | GADFA: Generator-Assisted Decision-Focused Approach for Opinion Expressing Timing Identification                                       | [1, 5, 13]                           |
| [154] | 1.20            | Learning Dynamic Multimodal Implicit and Explicit Networks for Multiple Financial Tasks                                                | [2, 3, 4]                            |

# References

- [1] [What You Say and How You Say It Matters: Predicting Stock Volatility Using Verbal and Vocal Cues](#)  
Yu Qin and Yi Yang. Unknown journal, 2019. 97 citations.  
100% Topic Match  
Demonstrates a multimodal (text + audio) deep model predicting short-horizon post-call stock volatility.  
Trains an RNN-based multimodal deep regression model on aligned earnings-call transcripts and Praat audio features to predict realized volatility at  $\Delta = 3, 7, 15, 30$  days; best gains at  $\Delta = 7$  versus text-only and past-volatility baselines.  
Uses S&P500 quarterly calls (576 calls, sentence-level GloVe-300 and 27-d acoustic features), evaluates realized-volatility (std. dev. of returns) horizons, reports notable out-of-sample MSE reductions but notes noisy alignment and limited sample size.
- [2] [HTML: Hierarchical Transformer-based Multi-task Learning for Volatility Prediction](#)  
Linyi Yang, ..., and Ruihai Dong. Proceedings of The Web Conference 2020, 2020. 120 citations.  
100% Topic Match  
Proposes a hierarchical transformer multi-task model using earnings-call text and audio to predict short- and long-horizon volatility.  
Implements a Hierarchical Transformer for transcripts + 27 Praat acoustic features, fuses modalities and jointly predicts log-volatility for  $n=\{3, 7, 15, 30\}$  days with large OOS gains.  
Relevant but not fully aligned: targets are  $n$  day realized/log-volatility from returns (emphasizes  $n=3$  as "short"), no mention of intraday/event-window alignment to call timing or any option/implicit-volatility targets.
- [3] [VolTAGE: Volatility Forecasting via Text-Audio Fusion with Graph Convolution Networks for Earnings Calls](#)  
Ramit Sawhney, ..., and R. Shah. Unknown journal, 2020. 43 citations.  
100% Topic Match  
Proposes a multimodal neural model (VolTAGE) to predict short-horizon stock volatility.  
Fuses FinBERT-encoded transcripts and aligned vocal features with inter-utterance attention, then applies a GCN over stocks in a semi-supervised multi-task regression for log realized volatility  $v[0, n]$  and  $v_n$ .  
Uses earnings-call audio+transcripts for S&P500 firms, emphasizes CEO vocal cues and cross-stock dependence; exact acoustic features, alignment details, and forecast horizon  $n$  are not fully specified.
- [4] [Multimodal Multi-Task Financial Risk Forecasting](#)  
Ramit Sawhney, ..., and Roger Zimmermann. Proceedings of the 28th ACM International Conference on Multimedia, 2020. 60 citations.  
100% Topic Match  
Demonstrates a deep multimodal (text+audio) model for predicting price movement and volatility.  
Trains a text-audio attention network with domain-specific text features and audio alignment in a multi-task ensemble for volatility and return prediction.  
Uses earnings-call multimodal inputs, but paper likely targets short-horizon price/volatility prediction generally — check exact volatility horizon/labels (realized vs implied, event-window timing) and whether out-of-sample forecasting setup (cross-sectional or time-series) and econometric controls are reported.
- [5] [An Empirical Investigation of Bias in the Multimodal Analysis of Financial Earnings Calls](#)  
Ramit Sawhney, ..., and R. Shah. Unknown journal, 2021. 20 citations.  
100% Topic Match  
Demonstrates a multimodal (text+audio) neural model that predicts short-horizon post-call volatility.  
Trains models on S&P500 earnings calls (transcripts + aligned audio) to predict average log realized volatility over 3, 7, 15, 30-day horizons; compares to text-only baselines.  
Relevant: uses call-level realized volatility as target and shows multimodal improves accuracy but introduces gender bias (female speakers underrepresented); dataset, code, and event-horizon definitions provided.
- [6] [ECC Analyzer: Extracting Trading Signal from Earnings Conference Calls using Large Language Model for Stock Volatility Prediction](#)  
Yupeng Cao, ..., and Papa Momar Ndiaye. Proceedings of the 5th ACM International Conference on AI in Finance, 2024. 12 citations.  
100% Topic Match  
Proposes a multimodal LLM-based framework (ECC Analyzer) for forecasting short-horizon stock volatility.  
Uses pretrained models to extract text and audio embeddings, LLM hierarchical extraction (paragraph summaries + RAG with a finance question bank), then fuses features to regress  $n$  day volatility.  
Evaluated on 572 aligned S&P500 calls (2017), improves 3 and 7 day forecasts vs GARCH/LSTM/multimodal baselines; paper omits clear label definition (realized vs implied), event window alignment, and whether predictions are cross sectional or firm time series.
- [7] [AMA-LSTM: Pioneering Robust and Fair Financial Audio Analysis for Stock Volatility Prediction](#)  
Shengkun Wang, ..., and Chang-Tien Lu. ArXiv, 2024. 4 citations.  
100% Topic Match  
Proposes an adversarially trained multimodal LSTM (AMA-LSTM) for stock volatility prediction.  
Combines FinBERT text embeddings and extracted audio cues, injecting adversarial perturbations to improve robustness and reduce gender bias; evaluated on two earnings-call datasets vs LSTM/SOTA.  
Missing key details for relevance: paper does not specify volatility target/horizon (realized vs implied, intraday/event-window), label alignment to call timing, cross-sectional vs time-series setup, or exact audio features and fusion architecture.
- [8] [DocFin: Multimodal Financial Prediction and Bias Mitigation using Semi-structured Documents](#)  
Puneet Mathur, ..., and Dinesh Manocha. Unknown journal, 2022. 4 citations.  
100% Topic Match  
Proposes a multimodal neural model fusing earnings-call audio, transcripts, and tabular financial documents.  
Implements audio prosodic features + transcript text + extracted tables; reports 5–12% improvement in volatility and price-movement prediction.  
Missing key details for your query: paper uses "stock volatility" but does not specify very-short horizons (intraday/T+1/event windows) nor whether volatility is realized vs. implied.
- [9] [RiskLabs: Predicting Financial Risk Using Large Language Model Based on Multi-Sources Data](#)  
Yupeng Cao, ..., and Papa Momar Ndiaye. ArXiv, 2024. 26 citations.  
100% Topic Match  
Proposes a multimodal LLM-augmented framework combining earnings-call audio and transcripts to predict volatility and VaR.  
Implements specialized encoders for calls, news, and time series; fuses LLM-derived transcript embeddings with acoustic embeddings and trains multi-task DL models (ablation compares Audio+Text).  
Relevant but only partially: uses multi-day horizons (3–30 days), no intraday/event-window or next-day RV labeling, and does not model option-implicit volatility.
- [10] [RiskLabs: Predicting Financial Risk Using Large Language Model based on Multimodal and Multi-Sources Data](#)  
Yupeng Cao, ..., and Haohang Li. ArXiv, 2024. 1 citations.  
100% Topic Match  
Proposes a multimodal LLM framework (RiskLabs) to predict financial risk including volatility.  
Uses earnings conference call transcripts + audio, pre-event price series, and news with LLM embeddings and multimodal fusion to forecast volatility/variance, VaR, and term-structured volatility.

Relevant but partially ambiguous: targets described as "volatility/variance" and short term ECC-induced volatility, yet paper lacks clear specification of realized vs implied IV, exact event horizons/window definitions, volatility construction, and whether predictions are cross-sectional, time-series, or panel.

- [11] [The Sound of Risk: A Multimodal Physics-Informed Acoustic Model for Forecasting Market Volatility and Enhancing Market Interpretability](#)  
Xiaoliang Chen, ..., and Jiaying Xie. ArXiv, 2025. 0 citations.  
100% Topic Match  
Proposes a multimodal (text + acoustic) model linking executive affect to post-call volatility.  
Maps acoustic and LLM-derived textual emotions into a 3-D ASL space (Tension, Stability, Arousal), engineers ~150 call-level features, and predicts realized volatility with XGBoost (bootstrap validation, ablations).  
Relevant? Uses true audio paralinguistic features and transcripts on 1,795 calls and achieves up to  $R^2=0.438$  for 30 day realized volatility, but does NOT study very short-horizon/event-window or implied volatility nor clearly specify cross-sectional vs time-series estimation.
- [12] [ECC Analyzer: Extract Trading Signal from Earnings Conference Calls using Large Language Model for Stock Performance Prediction](#)  
Yupeng Cao, ..., and Papa Momar Ndiaye. ArXiv, 2024. 7 citations.  
100% Topic Match  
Proposes a multimodal LLM-based pipeline to extract text and audio features from earnings calls.  
Uses paragraph-level LLM summaries, RAG with a finance question bank, pretrained audio/text embeddings, and multimodal fusion to predict stock volatility.  
Ambiguities: paper lacks specification of volatility type (realized vs implied), prediction horizon (intraday/next-day/event window), cross-sectional vs time-series setup, label construction, evaluation protocol, speaker diarization, and detailed acoustic/model architectures.
- [13] [MAEC: A Multimodal Aligned Earnings Conference Call Dataset for Financial Risk Prediction](#)  
Jiazheng Li, ..., and Ruihai Dong. Proceedings of the 29th ACM International Conference on Information & Knowledge Management, 2020. 59 citations.  
100% Topic Match  
Introduces a large-scale, multimodal earnings-call dataset (MAEC) with aligned text and audio.  
Builds sentence-level text-audio alignment for 3,443 calls (2015–2018, S&P1500), supplies linguistic features, acoustic features and MFCCs, and provides baseline financial risk/volatility prediction MSE results.  
Useful resource for ML/DL multimodal volatility work, but paper lacks clear short-horizon volatility target definitions, event-window alignment details, and full model architecture descriptions—so relevance depends on whether you will construct precise labels and prediction setups yourself.
- [14] [Risk Forecasting from Earnings Calls Acoustics and Network Correlations](#)  
Ramit Sawhney, ..., and R. Shah. Unknown journal, 2020. 7 citations.  
99% Topic Match  
Proposes a neural multimodal model using earnings-call audio and text to forecast stock risk/volatility.  
Combines cross inter-modal attention for verbal-vocal coherence and multi-layer network embeddings to capture stock interdependence; shows outperformance versus baselines.  
Relevant: uses audio acoustic features (not just transcripts) and joint text+audio ML, but check horizon/label details—paper abstracts "risk/volatility" forecasting but does not state very short-horizon realized/option-implied windows or exact prediction setup (cross-sectional vs time-series) in the provided summary.
- [15] [NumHTML: Numeric-Oriented Hierarchical Transformer Model for Multi-task Financial Forecasting](#)  
Linyi Yang, ..., and B. Smyth. Unknown journal, 2022. 48 citations.  
83% Topic Match  
Proposes a numeric-oriented hierarchical Transformer (NumHTML) for multimodal financial forecasting.  
Trains a multi-task model on aligned earnings-call text + Praat-extracted audio features, predicting returns with volatility as an auxiliary task.  
Uses multimodal earnings-call data and acoustic features, but paper does not specify volatility target (realized vs implied), horizon, or cross-sectional/time-series setup — relevance uncertain without those details.
- [16] [Enhancing Volatility Forecasting in Financial Markets: A General Numeral Attachment Dataset for Understanding Earnings Calls](#)  
Ming-Xuan Shi, ..., and Hsin-Hsi Chen. Unknown journal, 2023. 4 citations.  
48% Topic Match  
Proposes a numeral-attachment pretraining (GNA-Vol) to improve earnings-call understanding for volatility forecasting.  
Trains language models on an enhanced numeral-attachment dataset and shows improved MSE for 3 day and 7 day volatility forecasts versus text/audio baselines.  
Relevant but incomplete: uses short-horizon volatility (3/7-day) and compares multimodal baselines, yet paper does not clearly state realized vs implied volatility, precise event-window alignment to call timing, or whether predictions are cross-sectional, time series, or panel.
- [17] [Distilling Numeral Information for Volatility Forecasting](#)  
Chung-Chi Chen, ..., and Hsin-Hsi Chen. Proceedings of the 30th ACM International Conference on Information & Knowledge Management, 2021. 8 citations.  
15% Topic Match  
Demonstrates improved short-horizon volatility forecasting using numeral-aware text models.  
Trains a Numeral-Aware Model (NAM) on a new ECNum dataset of earnings-call transcripts, distilling numeric cues into embeddings and feeding them to volatility prediction models.  
Uses text-only inputs (transcripts) focusing on numeral extraction; does not use audio features and targets short-term realized volatility (event/short-horizon), so partially relevant (text multimodality and audio absent).
- [18] [Financial Risk Prediction with Multi-Round Q&A Attention Network](#)  
Zhen Ye, ..., and Wei Xu. Unknown journal, 2020. 23 citations.  
7% Topic Match  
Proposes a Multi-Round Q&A Attention Network to predict post-earnings-call volatility.  
Trains a transcript-only deep model that encodes dialogue rounds with bidirectional attention, then forecasts volatility after calls.  
Uses only textual transcripts (no audio), predicts short-horizon post-call volatility; evaluates out-of-sample performance versus baselines but paper unclear on exact volatility horizon (intraday/T+1/event-window) and whether realized or implied volatility.
- [19] [How Does Textual Information Selection Influence Time Series Forecasting? A Cross-modal Perspective on Financial Volatility Prediction](#)  
Hao Niu, ..., and Yao Zhang. 2024 IEEE International Conference on Multimedia and Expo (ICME), 2024. 0 citations.  
3% Topic Match  
Proposes a denoising multimodal model for volatility forecasting.  
Trains a Differentiable Binary Selector to pick helpful transcript sentences and a Volatility Predictor that integrates selected text into time-series forecasting.  
Uses earnings-call transcript text paired with time series volatility targets; paper focuses on text\*time-series interaction (no mention of audio, event-window or implied volatility specifics).
- [20] [Earnings Call Analysis Using a Sparse Attention Based Encoder and Multi-Source Counterfactual Augmentation](#)  
Zixuan Yuan, ..., and Hui Xiong. Proceedings of the Fourth ACM International Conference on AI in Finance, 2023. 4 citations.  
2% Topic Match  
Proposes a sparse-attention text encoder plus counterfactual augmentation for earnings-call analysis.  
Trains a text-only model (sparse self-attention) using unlabeled financial text as constrained perturbations to improve short- (3-day) and longer (15-day) volatility classification and sentiment tasks.  
Relevant but not fully aligned: predicts short-horizon volatility (3 and 15 days) from transcripts only — no audio/prosodic features, discretizes volatility into 3 classes rather than continuous RV/IV, and uses classification rather than direct real-valued variance forecasting.
- [21] [PRoFET: Predicting the Risk of Firms from Event Transcripts](#)  
Kilian Theil, ..., and H. Stuckenschmidt. Unknown journal, 2019. 20 citations.

## 2% Topic Match

Proposes a neural model (PRoFET) to predict firm risk/volatility from earnings-call transcripts. Combines attention-based semantic language representations with comprehensive financial features to forecast volatility, outperforming prior baselines. Uses transcripts only (textual modality); no audio/prosodic features reported; target is volatility but check horizon/event-window definitions (paper emphasizes quarterly recurring events).

### [22] [Less Is More: Volatility Forecasting with Contrastive Representation Learning \(Student Abstract\)](#)

Yanlong Huang, ..., and Kunpeng Zhang. Unknown journal, 2023. 1 citations.

## 1% Topic Match

Abstract: Earnings conference calls are indicative information events for volatility forecasting, which is essential for financial risk management and asset pricing. Although recent volatility forecasting models have explored the textual content of conference calls for prediction, they suffer from modeling the long-text and representing the risk-relevant information. This work proposes to identify key sentences for robust and interpretable transcript representation learning based on the cognitive theory. Specifically, we introduce TextRank to find key sentences and leverage attention mechanism to screen out the candidates by modeling the semantic correlations. Upon on the structural information of earning conference calls, we propose a structure-based...

### [23] [Exploring Hypergraph of Earnings Call for Risk Prediction \(Student Abstract\)](#)

Yi He, ..., and Yi Yang. Unknown journal, 2023. 1 citations.

## 1% Topic Match

Abstract: In financial economics, studies have shown that the textual content in the earnings conference call transcript has predictive power for a firm's future risk. However, the conference call transcript is very long and contains diverse non-relevant content, which poses challenges for the text-based risk forecast. This study investigates the structural dependency within a conference call transcript by explicitly modeling the dialogue between managers and analysts. Specifically, we utilize TextRank to extract information and exploit the semantic correlation within a discussion using hypergraph learning. This novel design can improve the transcript representation performance and reduce the risk of forecast errors. Experimental...

### [24] [Convolutional Neural Networks for Financial Text Regression](#)

Nesat Dereli and M. Saraçlar. Unknown journal, 2019. 17 citations.

## 1% Topic Match

Abstract: Forecasting financial volatility of a publicly-traded company from its annual reports has been previously defined as a text regression problem. Recent studies use a manually labeled lexicon to filter the annual reports by keeping sentiment words only. In order to remove the lexicon dependency without decreasing the performance, we replace bag-of-words model word features by word embedding vectors. Using word vectors increases the number of parameters. Considering the increase in number of parameters and excessive lengths of annual reports, a convolutional neural network model is proposed and transfer learning is applied. Experimental results show that the convolutional neural network model...

### [25] [Predicting Realized Variance Out of Sample: Can Anything Beat The Benchmark?](#)

Austin Pollok. ArXiv, 2025. 0 citations.

## 1% Topic Match

Abstract: The discrepancy between realized volatility and the market's view of volatility has been known to predict individual equity options at the monthly horizon. It is not clear how this predictability depends on a forecast's ability to predict firm-level volatility. We consider this phenomenon at the daily frequency using high-dimensional machine learning models, as well as low-dimensional factor models. We find that marginal improvements to standard forecast error measurements can lead to economically significant gains in portfolio performance. This makes a case for re-imagining the way we train models that are used to construct portfolios.

### [26] [The Impact of Differential Privacy on Group Disparity Mitigation](#)

Victor Petrén Bach Hansen, ..., and Anders Søgaard. ArXiv, 2022. 12 citations.

## 1% Topic Match

Abstract: The performance cost of differential privacy has, for some applications, been shown to be higher for minority groups fairness, conversely, has been shown to disproportionately compromise the privacy of members of such groups. Most work in this area has been restricted to computer vision and risk assessment. In this paper, we evaluate the impact of differential privacy on fairness across four tasks, focusing on how attempts to mitigate privacy violations and between-group performance differences interact Does privacy inhibit attempts to ensure fairness? To this end, we train epsilon, delta-differentially private models with empirical risk minimization and group distributionally robust training...

### [27] [DialogueGAT: A Graph Attention Network for Financial Risk Prediction by Modeling the Dialogues in Earnings Conference Calls](#)

Yunxin Sang and Yangming Bao. Unknown journal, 2022. 9 citations.

## 1% Topic Match

Abstract: ,

### [28] [Multi-Domain Transformer-Based Counterfactual Augmentation for Earnings Call Analysis](#)

Zixuan Yuan, ..., and Hui Xiong. ArXiv, 2021. 2 citations.

## 1% Topic Match

Abstract: Earnings call (EC), as a periodic teleconference of a publicly-traded company, has been extensively studied as an essential market indicator because of its high analytical value in corporate fundamentals. The recent emergence of deep learning techniques has shown great promise in creating automated pipelines to benefit the EC-supported financial applications. However, these methods presume all included contents to be informative without refining valuable semantics from long-text transcript and suffer from EC scarcity issue. Meanwhile, these black-box methods possess inherent difficulties in providing human-understandable explanations. To this end, in this paper, we propose a Multi-Domain Transformer-Based Counterfactual Augmentation, named MTCA, to...

### [29] [Predicting Abnormal Stock Return Volatility Using Textual Analysis of News ? A Meta-Learning Approach](#)

Ren'ita Mykov!, ..., and V. OlejThe AMFITEATRU ECONOMIC journal, 2018. 8 citations.

## 1% Topic Match

Abstract: Textual analysis of news articles is increasingly important in predicting stock prices. Previous research has intensively utilized the textual analysis of news and other firm-related documents in volatility prediction models. It has been demonstrated that the news may be related to abnormal stock price behavior subsequent to their dissemination. However, previous studies to date have tended to focus on linear regression methods in predicting volatility. Here, we show that non-linear models can be effectively employed to explain the residual variance of the stock price. Moreover, we use meta-learning approach to simulate the decision-making process of various investors. The results suggest...

### [30] [Stock Volatility Prediction Based on Self-attention Networks with Social Information](#)

Jie Zheng, ..., and Zengchang Qin. 2019 IEEE Conference on Computational Intelligence for Financial Engineering & Economics (CIFEr), 2019. 17 citations.

## 0% Topic Match

Abstract: Stock volatility prediction is a challenging task in time-series prediction according to the Efficient Market Hypothesis which supposes all the investors are rational. However, many theories have showed that stock markets are not efficient due to the effects of psychological and social factors. In this paper, we constructed self-attention networks (SAN) to quantify the impact on the volatility of Chinese stock market of social information, such as social opinion and social concern. Our SAN model can explore the relationships among features at different time steps more flexibly, and thus, explore stock historical information more effectively. Empirical results show the superiority...

### [31] [KeFVP: Knowledge-enhanced Financial Volatility Prediction](#)

Hao Niu, ..., and Weizu Yang. Unknown journal, 2023. 7 citations.

## 0% Topic Match

Abstract: .

### [32] [Predicting Corporate Risk by Jointly Modeling Company Networks and Dialogues in Earnings Conference Calls](#)

Yu Sang and Yang Bao. ArXiv, 2022. 2 citations.

## 0% Topic Match

Abstract: Earnings conference calls are significant information events for volatility forecasting, which is essential for financial risk management and asset pricing. Although some recent volatility forecasting models have utilized the textual content of conference calls, the dialogue structures of conference calls and company relationships are almost ignored in extant literature. To bridge this gap, we propose a new model called Temporal Virtual Graph Neural Network (TVGNN) for volatility forecasting by jointly modeling conference call dialogues and company networks. Our model differs from existing models in several important ways. First, we propose to exploit more dialogue structures by encoding position, utterance, speaker...

[33] [Stock Price Volatility Prediction: A Case Study with AutoML](#)

Hilal Pataci, ..., and Lucian Popa. Proceedings of the Fourth Workshop on Financial Technology and Natural Language Processing (FinNLP), 2022. 5 citations.

0% Topic Match

Abstract: Accurate prediction of the stock price volatility, the rate at which the price of a stock increases or decreases over a particular period, is an important problem in finance. Inaccurate prediction of stock price volatility might lead to investment risk and financial loss, while accurate prediction might generate significant returns for investors. Several studies investigated stock price volatility prediction in a regression task by using the transcripts of earning calls (quarterly conference calls held by public companies) with Natural Language Processing (NLP) techniques. Existing studies use the entire transcript and this degrades the performance due to noise caused by irrelevant...

[34] [Divide and Contrast: A Text-Based Method for Firm Market Risk Prediction](#)

Yi He, ..., and Kunpeng Zhang. INFORMS Journal on Computing, 2025. 0 citations.

0% Topic Match

Abstract: Forecasting the market risk for publicly traded companies is a critical task for market participants. Financial economics research demonstrates that the textual information contained in corporate disclosures, such as earnings conference call transcripts, can effectively predict a firm's future risk. This finding has inspired a growing body of research focused specifically on transcript-based approaches to risk forecasting. However, earnings transcripts are typically long documents with thousands of words. Prior transcript-based risk forecasting studies that represent the entire transcript as one text sequence often fail to capture risk-relevant information and fall short in risk forecasting. In this work, we propose a...

[35] [Multimodal Sentiment-Aware Volatility Forecasting: A Deep Fusion Approach with Social Media Data](#)

Chenxi Wang, ..., and Qiuyi Jiang. Proceedings of the 2025 International Conference on Generative Artificial Intelligence for Business, 2025. 0 citations.

0% Topic Match

Abstract: This paper proposes an innovative multimodal emotion perception framework to improve the accuracy of financial volatility prediction by fusing social media graphic data. Traditional studies mainly rely on a single textual modality, which makes it difficult to capture the implicit emotions in visual information and the scarcity of labelled data in the financial domain. In this paper, we propose three core innovations: combining social media text (e.g., comments) and images (e.g., emoticons, news screenshots) to achieve adaptive fusion of graphical and textual features through a cross-modal attention mechanism to improve the completeness of emotion representation; adopting a two-stage strategy of...

[36] [Utilization of deep learning to mine insights from earning calls for stock price movement predictions](#)

Zhiqiang Ma, ..., and Xiaomo Liu. Proceedings of the First ACM International Conference on AI in Finance, 2020. 2 citations.

0% Topic Match

Abstract: Earnings calls are hosted by management of public companies to discuss the company's financial performance with analysts and investors. Information disclosed from an earning call is an essential source of data for analysts and investors to make investment decisions. Thus, we leverage earning call transcripts combined with companies' historical stock data and sector information to predict company's stock price movements. We propose to model these three features in a deep learning framework jointly, where attention mechanism is applied to the earnings call textual feature and a recurrent neural network (RNN) is used on the sequential stock price data. Our empirical...

[37] [Speech emotion recognition and text sentiment analysis for financial distress prediction](#)

P. Hájek and Michal Munk. Neural Computing and Applications, 2023. 36 citations.

0% Topic Match

Abstract: In recent years, there has been an increasing interest in text sentiment analysis and speech emotion recognition in finance due to their potential to capture the intentions and opinions of corporate stakeholders, such as managers and investors. A considerable performance improvement in forecasting company financial performance was achieved by taking textual sentiment into account. However, far too little attention has been paid to managerial emotional states and their potential contribution to financial distress prediction. This study seeks to address this problem by proposing a deep learning architecture that uniquely combines managerial emotional states extracted using speech emotion recognition with FinBERT-based...

[38] [Same Company, Same Signal: The Role of Identity in Earnings Call Transcripts](#)

Ding Yu, ..., and Hangfeng He. Unknown journal, 2024. 0 citations.

0% Topic Match

Abstract: Post-earnings volatility prediction is critical for investors, with previous works often leveraging earnings call transcripts under the assumption that their rich semantics contribute significantly. To further investigate how transcripts impact volatility, we introduce DEC, a dataset featuring accurate volatility calculations enabled by the previously overlooked beforeAfterMarket attribute and dense ticker coverage. Unlike established benchmarks, where each ticker has only around two earnings, DEC provides 20 earnings records per ticker. Using DEC, we reveal that post-earnings volatility undergoes significant shifts, with each ticker displaying a distinct volatility distribution. To leverage historical post-earnings volatility and capture ticker-specific patterns, we propose two training-free...

[39] [Causal Inference from Text: Unveiling Interactions between Variables](#)

Yuxiang Zhou and Yulan He. Unknown journal, 2023. 5 citations.

0% Topic Match

Abstract: Adjusting for latent covariates is crucial for estimating causal effects from observational textual data. Most existing methods only account for confounding covariates that affect both treatment and outcome, potentially leading to biased causal effects. This bias arises from insufficient consideration of non-confounding covariates, which are relevant only to either the treatment or the outcome. In this work, we aim to mitigate the bias by unveiling interactions between different variables to disentangle the non-confounding covariates when estimating causal effects from text. The disentangling process ensures covariates only contribute to their respective objectives, enabling independence between variables. Additionally, we impose a constraint...

[40] [Volatility Forecasting with 1-dimensional CNNs via transfer learning](#)

Bernadett Aradi, ..., and J'ozsef G'all. arXiv: Statistical Finance, 2020. 0 citations.

0% Topic Match

Abstract: Volatility is a natural risk measure in finance as it quantifies the variation of stock prices. A frequently considered problem in mathematical finance is to forecast different estimates of volatility. What makes it promising to use deep learning methods for the prediction of volatility is the fact, that stock price returns satisfy some common properties, referred to as 'stylized facts'. Also, the amount of data used can be high, favoring the application of neural networks. We used 10 years of daily prices for hundreds of frequently traded stocks, and compared different CNN architectures: some networks use only the considered stock,...

[41] [Predicting Stock Volatility from Quarterly Earnings Calls and Transcript Summaries using Text Regression](#)

Naveed Ahmad and Aram Zinzalian. Journal Not Provided, 2010. 3 citations.

0% Topic Match

No summary or abstract available

[42] [Harnessing Earnings Reports for Stock Predictions: A QLoRA-Enhanced LLM Approach](#)

Haowei Ni, ..., and Yuxi Chan. 2024 6th International Conference on Data-driven Optimization of Complex Systems (DOCS), 2024. 44 citations.

0% Topic Match

Abstract: Accurate stock market predictions following earnings reports are crucial for investors. Traditional methods, particularly classical machine learning models, struggle with these predictions because they cannot effectively process and interpret extensive textual data contained in earnings reports and often overlook nuances that influence market movements. This paper introduces an advanced approach by employing Large Language Models (LLMs) instruction fine-tuned with a novel combination of instruction-based techniques and quantized low-rank adaptation (QLoRA) compression. Our methodology integrates 'base factors', such as financial metric growth and earnings transcripts, with 'external factors', including recent market indices performances and analyst grades, to create a rich, supervised...

[43] [ECHO-GL: Earnings Calls-Driven Heterogeneous Graph Learning for Stock Movement Prediction](#)

Mengpu Liu, ..., and Xiaolin Zheng. Unknown journal, 2024. 14 citations.

0% Topic Match

Abstract: Stock movement prediction serves an important role in quantitative trading. Despite advances in existing models that enhance stock movement prediction by incorporating stock relations, these prediction models face two limitations, i.e., constructing either insufficient or static stock relations, which fail to effectively capture the complex dynamic stock relations because such complex dynamic stock relations are influenced by various factors in the ever-changing financial market. To tackle the above limitations, we propose a novel stock movement prediction model ECHO-GL based on stock relations derived from earnings calls. ECHO-GL not only constructs comprehensive stock relations by exploiting the rich semantic information in...

[44] [News Implied Volatility and Disaster Concerns](#)

Asaf Manela and Alan Moreira. ERN: Forecasting & Simulation (Prices) (Topic), 2015. 507 citations.

0% Topic Match

No summary or abstract available

[45] [Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models](#)

Ha Young Kim and C. H. Won. Expert Syst. Appl., 2018. 641 citations.

0% Topic Match

No summary or abstract available

[46] [Multimodal Multi-Speaker Merger & Acquisition Financial Modeling: A New Task, Dataset, and Neural Baselines](#)

Ramit Sawhney, ..., and R. Shah. Unknown journal, 2021. 15 citations.

0% Topic Match

Abstract: Risk prediction is an essential task in financial markets. Merger and Acquisition (M&A) calls provide key insights into the claims made by company executives about the restructuring of the financial firms. Extracting vocal and textual cues from M&A calls can help model the risk associated with such financial activities. To aid the analysis of M&A calls, we curate a dataset of conference call transcripts and their corresponding audio recordings for the time period ranging from 2016 to 2020. We introduce M3ANet, a baseline architecture that takes advantage of the multimodal multi-speaker input to forecast the financial risk associated with the...

[47] [Using Structured Events to Predict Stock Price Movement: An Empirical Investigation](#)

Xiao Ding, ..., and Junwen Duan. Unknown journal, 2014. 247 citations.

0% Topic Match

Abstract: It has been shown that news events influence the trends of stock price movements. However, previous work on news-driven stock market prediction rely on shallow features (such as bags-of-words, named entities and noun phrases), which do not capture structured entity-relation information, and hence cannot represent complete and exact events. Recent advances in Open Information Extraction (Open IE) techniques enable the extraction of structured events from web-scale data. We propose to adapt Open IE technology for event-based stock price movement prediction, extracting structured events from large-scale public news without manual efforts. Both linear and nonlinear models are employed to empirically investigate...

[48] [FinCall-Surprise: A Large Scale Multi-modal Benchmark for Earning Surprise Prediction](#)

Dong Shu, ..., and Mengnan Du. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Predicting corporate earnings surprises is a profitable yet challenging task, as accurate forecasts can inform significant investment decisions. However, progress in this domain has been constrained by a reliance on expensive, proprietary, and text-only data, limiting the development of advanced models. To address this gap, we introduce `FinCall-Surprise` (Financial Conference Call for Earning Surprise Prediction), the first large-scale, open-source, and multi-modal dataset for earnings surprise prediction. Comprising 2,688 unique corporate conference calls from 2019 to 2021, our dataset features word-to-word conference call textual transcripts, full audio recordings, and corresponding presentation slides. We establish a comprehensive benchmark by evaluating 26 state-of-the-art...

[49] [Multimodal deep learning for short-term stock volatility prediction](#)

M. Sardelich and S. Manandhar. ArXiv, 2018. 30 citations.

0% Topic Match

Abstract: Stock market volatility forecasting is a task relevant to assessing market risk. We investigate the interaction between news and prices for the one-day-ahead volatility prediction using state-of-the-art deep learning approaches. The proposed models are trained either end-to-end or using sentence encoders transferred from other tasks. We evaluate a broad range of stock market sectors, namely Consumer Staples, Energy, Utilities, Healthcare, and Financials. Our experimental results show that adding news improves the volatility forecasting as compared to the mainstream models that rely only on price data. In particular, our model outperforms the widely-recognized GARCH(1,1) model for all sectors in terms of...

[50] [Word Embeddings-Based Uncertainty Detection in Financial Disclosures](#)

Kilian Theil, ..., and H. Stuckenschmidt. Unknown journal, 2018. 22 citations.

0% Topic Match

Abstract: In this paper, we use NLP techniques to detect linguistic uncertainty in financial disclosures. Leveraging general-domain and domain-specific word embedding models, we automatically expand an existing dictionary of uncertainty triggers. We furthermore examine how an expert filtering affects the quality of such an expansion. We show that the dictionary expansions significantly improve regressions on stock return volatility. Lastly, we prove that the expansions significantly boost the automatic detection of uncertain sentences.

[51] [Exploiting the errors: A simple approach for improved volatility forecasting](#)

Tim Bollerslev, ..., and R. Quaadvlieg. Journal of Econometrics, 2016. 395 citations.

0% Topic Match

No summary or abstract available

[52] [Explainable Text-Driven Neural Network for Stock Prediction](#)

Linyi Yang, ..., and Ruihai Dong. 2018 5th IEEE International Conference on Cloud Computing and Intelligence Systems (CCIS), 2018. 53 citations.

0% Topic Match

Abstract: It has been shown that financial news leads to the fluctuation of stock prices. However, previous work on news-driven financial market prediction focused only on predicting stock price movement without providing an explanation. In this paper, we propose a dual-layer attention-based neural network to address this issue. In the initial stage, we introduce a knowledge-based method to adaptively extract relevant financial news. Then, we use an input attention to pay more attention to the more influential news and concatenate the day embeddings with the output of the news representation. Finally, we use an output attention mechanism to allocate different weights...

[53] [Analyzing Firm Reports for Volatility Prediction: A Knowledge-Driven Text-Embedding Approach](#)

Yi Yang, ..., and Yangyang Fan. INFORMS J. Comput., 2021. 24 citations.

0% Topic Match

Abstract: Predicting stock return volatility is the key to investment and risk management. Traditional volatility-forecasting methods primarily rely on stochastic models. More recently, many machine-learning approaches, particularly text-mining techniques, have been implemented to predict stock return volatility, thus taking

advantage of the availability of large amounts of unstructured data such as firm financial reports. Most existing studies develop simple but effective models to analyze text, such as dictionary-based matching algorithms that use a set of manually constructed keywords. However, the latent and deep semantics encoded in text are usually neglected. In this study, we build on recent progress in representation learning...

[54] [Modal-adaptive Knowledge-enhanced Graph-based Financial Prediction from Monetary Policy Conference Calls with LLM](#)

Kun Ouyang, ..., and Xu Sun. ArXiv, 2024. 14 citations.

0% Topic Match

Abstract: Financial prediction from Monetary Policy Conference (MPC) calls is a new yet challenging task, which targets at predicting the price movement and volatility for specific financial assets by analyzing multimodal information including text, video, and audio. Although the existing work has achieved great success using cross-modal transformer blocks, it overlooks the potential external financial knowledge, the varying contributions of different modalities to financial prediction, as well as the innate relations among different financial assets. To tackle these limitations, we propose a novel Modal-Adaptive Knowledge-enhanced Graph-based Financial Prediction scheme, named MANAGER. Specifically, MANAGER resorts to FinDKG to obtain the external related...

[55] [Stock Movement and Volatility Prediction from Tweets, Macroeconomic Factors and Historical Prices](#)

Shengkun Wang, ..., and Chang-Tien Lu. 2023 IEEE International Conference on Big Data (BigData), 2023. 5 citations.

0% Topic Match

Abstract: Predicting stock market is vital for investors and policymakers, acting as a barometer of the economic health. We leverage social media data, a potent source of public sentiment, in tandem with macroeconomic indicators as government-compiled statistics, to refine stock market predictions. However, prior research using tweet data for stock market prediction faces three challenges. First, the quality of tweets varies widely. While many are filled with noise and irrelevant details, only a few genuinely mirror the actual market scenario. Second, solely focusing on the historical data of a particular stock without considering its sector can lead to oversight. Stocks within...

[56] [A Continuous Volatility Forecasting Model Based on Neural Differential Equations and Scale-Similarity](#)

Bowen Pang, ..., and Wei Wei. IEEE Transactions on Neural Networks and Learning Systems, 2024. 1 citations.

0% Topic Match

Abstract: Volatility forecasting is a problem in finance that attracts the attention of both academia and industry. While existing approaches typically utilize a discrete-time latent process that governs the volatility to forecast its future level, volatility is considered to evolve continuously, which makes discrete-time modeling inevitably lose some critical information about the evolution of volatility. In this article, a novel neural-network-based model, Continuous Volatility Forecasting Model, CVFM is proposed to tackle this problem. First, CVFM introduces a continuous-time latent process, whose evolution is modeled with neural differential equations (NDEs), to govern volatility, which effectively captures the continuous evolutionary behavior of volatility...

[57] [Forecasting Volatility Using Historical Data](#)

Stephen Figlewski. Risk Management, 1994. 63 citations.

0% Topic Match

No summary or abstract available

[58] [Financial Keyword Expansion via Continuous Word Vector Representations](#)

Ming-Feng Tsai and Chuan-Ju Wang. Unknown journal, 2014. 30 citations.

0% Topic Match

Abstract: This paper proposes to apply the continuous vector representations of words for discovering keywords from a financial sentiment lexicon. In order to capture more keywords, we also incorporate syntactic information into the Continuous Bag-of-Words (CBOW) model. Experimental results on a task of financial risk prediction using the discovered keywords demonstrate that the proposed approach is good at predicting financial risk.

[59] [A Semiparametric Gaussian Copula Regression Model for Predicting Financial Risks from Earnings Calls](#)

William Yang Wang and Zhenhao Hua. Unknown journal, 2014. 64 citations.

0% Topic Match

Abstract: Earnings call summarizes the financial performance of a company, and it is an important indicator of the future financial risks of the company. We quantitatively study how earnings calls are correlated with the financial risks, with a special focus on the financial crisis of 2009. In particular, we perform a text regression task: given the transcript of an earnings call, we predict the volatility of stock prices from the week after the call is made. We propose the use of copula: a powerful statistical framework that separately models the uniform marginals and their complex multivariate stochastic dependencies, while not requiring...

[60] [Deep Attentive Learning for Stock Movement Prediction from Social Media Text and Company Correlations](#)

Ramit Sawhney, ..., and R. Shah. Unknown journal, 2020. 155 citations.

0% Topic Match

Abstract: In the financial domain, risk modeling and profit generation heavily rely on the sophisticated and intricate stock movement prediction task. Stock forecasting is complex, given the stochastic dynamics and non-stationary behavior of the market. Stock movements are influenced by varied factors beyond the conventionally studied historical prices, such as social media and correlations among stocks. The rising ubiquity of online content and knowledge mandates an exploration of models that factor in such multimodal signals for accurate stock forecasting. We introduce an architecture that achieves a potent blend of chaotic temporal signals from financial data, social media, and inter-stock relationships via...

[61] [MONOPOLY: Financial Prediction from MONetary POLiCy Conference Videos Using Multimodal Cues](#)

Puneet Mathur, ..., and Dinesh Manocha. Proceedings of the 30th ACM International Conference on Multimedia, 2022. 20 citations.

0% Topic Match

Abstract: Risk prediction and price movement classification are essential tasks in financial markets. Monetary policy calls (MPC) provide important insights into the actions taken by a country's central bank on economic goals related to inflation, employment, prices, and interest rates. Analyzing visual, vocal, and textual cues from MPC calls can help analysts and policymakers evaluate the economic risks and make sound investment decisions. To aid the analysis of MPC calls, we curate the Monopoly dataset, a collection of public conference call videos along with their corresponding audio recordings and text transcripts released by six international banks between 2009 and 2022. Our...

[62] [Examining the Volatility of Taiwan Stock Index Returns via a Three-Volatility-Regime Markov-Switching ARCH Model](#)

M. Li and H. Lin. Review of Quantitative Finance and Accounting, 2003. 27 citations.

0% Topic Match

No summary or abstract available

[63] [Benchmarking Deep Sequential Models on Volatility Predictions for Financial Time Series](#)

Qiang Zhang, ..., and Y. Liu. ArXiv, 2018. 9 citations.

0% Topic Match

Abstract: Volatility is a quantity of measurement for the price movements of stocks or options which indicates the uncertainty within financial markets. As an indicator of the level of risk or the degree of variation, volatility is important to analyse the financial market, and it is taken into consideration in various decision-making processes in financial activities. On the other hand, recent advancement in deep learning techniques has shown strong capabilities in modelling sequential data, such as speech and natural language. In this paper, we empirically study the applicability of the latest deep structures with respect to the volatility modelling problem, through...

[64] [Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation](#)

R. Engle. Econometrica, 1982. 22163 citations.

0% Topic Match

No summary or abstract available

[65] [FinTech Applications](#)

Chung-Chi Chen, ..., and Hsin-Hsi Chen. From Opinion Mining to Financial Argument Mining, 2021. 2 citations.

0% Topic Match

Abstract: The Financial Stability Board defines financial technology (FinTech) as "technology-enabled innovation in financial services." At the 2015 World Economic Forum, experts proposed a taxonomy for financial services that can be classified into six major categories: payments, deposits and lending, market provisioning, capital raising, insurance, and investment management. Because financial opinion mining can be applied to many listed services, we survey various cases in this chapter and show that financial opinion mining is useful and crucial in many financial application scenarios.

[66] [Modeling Financial Analysts' Decision Making via the Pragmatics and Semantics of Earnings Calls](#)

Katherine A. Keith and Amanda Stent. Unknown journal, 2019. 59 citations.

0% Topic Match

Abstract: Every fiscal quarter, companies hold earnings calls in which company executives respond to questions from analysts. After these calls, analysts often change their price target recommendations, which are used in equity re- search reports to help investors make deci- sions. In this paper, we examine analysts' decision making behavior as it pertains to the language content of earnings calls. We identify a set of 20 pragmatic features of analysts' questions which we correlate with analysts' pre-call investor recommendations. We also analyze the degree to which semantic and pragmatic features from an earnings call complement market data in predicting analysts' post-call...

[67] [Option Pricing when the Variance Changes Randomly: Theory, Estimation, and an Application](#)

Louis O. Scott. Journal of Financial and Quantitative Analysis, 1987. 1119 citations.

0% Topic Match

No summary or abstract available

[68] [Multimodal Proposal for an AI-Based Tool to Increase Cross-Assessment of Messages](#)

Alejandro 'Alvarez Castro and Joaquín B. Ordieres Meré. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Earnings calls represent a uniquely rich and semistructured source of financial communication, blending scripted managerial commentary with unscripted analyst dialogue. Although recent advances in financial sentiment analysis have integrated multimodal signals, such as textual content and vocal tone, most systems rely on flat document-level or sentence-level models, failing to capture the layered discourse structure of these interactions. This paper introduces a novel multimodal framework designed to generate semantically rich and structurally aware embeddings of earnings calls, by encoding them as hierarchical discourse trees. Each node, comprising either a monologue or a question-answer pair, is enriched with emotional signals derived from...

[69] [Enhancing Prediction by Incorporating Entropy Loss in Volatility Forecasting](#)

R. Urniežius, ..., and Darius Ezerskis. Entropy, 2025. 0 citations.

0% Topic Match

Abstract: In this paper, we propose examining Heterogeneous Autoregressive (HAR) models using five different estimation techniques and four different estimation horizons to decide which performs better in terms of forecasting accuracy. Several different estimators are used to determine the coefficients of three selected HAR-type models. Furthermore, model lags, calculated using 5 min intraday data from the Standard & Poor's 500 (SPX) index and the Chicago Board Options Exchange Volatility (VIX) index as the sole exogenous variable, enrich the models. For comparison and evaluation of the experimental results, we use three metrics: Quasi-Likelihood (QLIKE), Mean Absolute Error (MAE), and Mean Squared Error...

[70] [Computing the Constant Elasticity of Variance Option Pricing Formula](#)

Mark Schroder. Journal of Finance, 1989. 347 citations.

0% Topic Match

No summary or abstract available

[71] [Stock Movement Prediction from Tweets and Historical Prices](#)

Yumo Xu and Shay B. Cohen. Unknown journal, 2018. 391 citations.

0% Topic Match

Abstract: Stock movement prediction is a challenging problem: the market is highly stochastic, and we make temporally-dependent predictions from chaotic data. We treat these three complexities and present a novel deep generative model jointly exploiting text and price signals for this task. Unlike the case with discriminative or topic modeling, our model introduces recurrent, continuous latent variables for a better treatment of stochasticity, and uses neural variational inference to address the intractable posterior inference. We also provide a hybrid objective with temporal auxiliary to flexibly capture predictive dependencies. We demonstrate the state-of-the-art performance of our proposed model on a new stock...

[72] [GADFA: Generator-Assisted Decision-Focused Approach for Opinion Expressing Timing Identification](#)

Chung-Chi Chen, ..., and Yusuke Miyao. ArXiv, 2024. 0 citations.

0% Topic Match

Abstract: The advancement of text generation models has granted us the capability to produce coherent and convincing text on demand. Yet, in real-life circumstances, individuals do not continuously generate text or voice their opinions. For instance, consumers pen product reviews after weighing the merits and demerits of a product, and professional analysts issue reports following significant news releases. In essence, opinion expression is typically prompted by particular reasons or signals. Despite long-standing developments in opinion mining, the appropriate timing for expressing an opinion remains largely unexplored. To address this deficit, our study introduces an innovative task - the identification of news-triggered...

[73] [Option Pricing when the Variance Is Changing](#)

H. Johnson and D. Shanno. Journal of Financial and Quantitative Analysis, 1987. 441 citations.

0% Topic Match

No summary or abstract available

[74] [Measuring Consistency in Text-based Financial Forecasting Models](#)

Linyi Yang, ..., and Yue Zhang. ArXiv, 2023. 4 citations.

0% Topic Match

Abstract: Financial forecasting has been an important and active area of machine learning research, as even the most modest advantages in predictive accuracy can be parlayed into significant financial gains. Recent advances in natural language processing (NLP) bring the opportunity to leverage textual data, such as earnings reports of publicly traded companies, to predict the return rate for an asset. However, when dealing with such a sensitive task, the consistency of models – their invariance under meaning-preserving alternations in input – is a crucial property for building user trust. Despite this, current methods for financial forecasting do not take consistency into...

[75] [Unlocking the Power of Voice for Financial Risk Prediction: A Theory-Driven Deep Learning Design Approach](#)

Yi Yang, ..., and Zhongju Zhang. MIS Q., 2023. 16 citations.

0% Topic Match

Abstract: Unstructured multimedia data (text and audio) provides unprecedented opportunities to derive actionable decision-making in the financial industry, in areas such as portfolio and risk management. However, due to formidable methodological challenges, the promise of business value from unstructured multimedia data has not materialized. In this study, we use a design science approach to develop DeepVoice, a novel nonverbal predictive analysis system for financial risk prediction, in the setting of quarterly earnings conference calls. DeepVoice forecasts financial risk by leveraging not only what managers say (verbal linguistic cues) but also how managers say it (vocal cues) during the earnings conference calls....

[76] [Term-Driven Forward-Looking Claim Synthesis in Earnings Calls](#)

Chung-Chi Chen and Hiroya Takamura. Unknown journal, 2024. 3 citations.

0% Topic Match

No summary or abstract available

[77] [A Prediction Approach for Stock Market Volatility Based on Time Series Data](#)

Sheikh Mohammad Idrees, ..., and Parul Agarwal. IEEE Access, 2019. 202 citations.

0% Topic Match

Abstract: Time series analysis and forecasting is of vital significance, owing to its widespread use in various practical domains. Time series data refers to an ordered sequence or a set of data points that a variable takes at equal time intervals. The stock market is considered to be one of the most highly complex financial systems which consist of various components or stocks, the price of which fluctuates greatly with respect to time. Stock market forecasting involves uncovering the market trends with respect to time. All the stock market investors aim to maximize the returns over their investments and minimize the...

[78] [Retrieving Implicit Information for Stock Movement Prediction](#)

Tsun-Hsien Tang, ..., and Hsin-Hsi Chen. Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021. 4 citations.

0% Topic Match

Abstract: Previous studies on the financial news focus mainly on the news articles explicitly mentioning the target financial instruments, and may suffer from data sparsity. As taking into consideration other related news, e.g., sector-related news, is a crucial part of real-world decision-making, we explore the use of news without explicit target mentions to enrich the information for the prediction model. We develop a neural network framework that jointly learns with a news selection mechanism to extract implicit information from the chaotic daily news pool. Our proposed model, called the news distilling network (NDN), takes advantage of neural representation learning and collaborative...

[79] [Stock Embeddings: Learning Distributed Representations for Financial Assets](#)

Rian Dolphin, ..., and Ruihai Dong. ArXiv, 2022. 9 citations.

0% Topic Match

Abstract: Identifying meaningful relationships between the price movements of financial assets is a challenging but important problem in a variety of financial applications. However with recent research, particularly those using machine learning and deep learning techniques, focused mostly on price forecasting, the literature investigating the modelling of asset correlations has lagged somewhat. To address this, inspired by recent successes in natural language processing, we propose a neural model for training stock embeddings, which harnesses the dynamics of historical returns data in order to learn the nuanced relationships that exist between financial assets. We describe our approach in detail and discuss a...

[80] [Earnings Guidance and Market Uncertainty](#)

Jonathan L. Rogers, ..., and Andrew Van Buskirk. Chicago Booth ARC: Financial Accounting (Topic), 2009. 301 citations.

0% Topic Match

No summary or abstract available

[81] [Numerals in Financial Narratives](#)

Chung-Chi Chen, ..., and Hsin-Hsi Chen. From Opinion Mining to Financial Argument Mining, 2021. 2 citations.

0% Topic Match

Abstract: Numerals are more common in financial narratives than in documents from other domains, which makes understanding numerals very important when analyzing financial documents. In this chapter, we summarize our work on numerals in financial narratives and share findings from the FinNum shared task series in the 14th and 15th NTCIR Conferences. In Sect. 5.1, we discuss how to understand the meaning of a given numeral, and in Sect. 5.2, we discuss numeral attachment, where we link numerals and named entities. In Sect. 5.3, we show experimental results from downstream tasks that demonstrate the importance of numeral understanding in financial narratives...

[82] [MiMIC: Multi-Modal Indian Earnings Calls Dataset to Predict Stock Prices](#)

Sohom Ghosh, ..., and Sudipta Naskar. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Predicting stock market prices following corporate earnings calls remains a significant challenge for investors and researchers alike, requiring innovative approaches that can process diverse information sources. This study investigates the impact of corporate earnings calls on stock prices by introducing a multi-modal predictive model. We leverage textual data from earnings call transcripts, along with images and tables from accompanying presentations, to forecast stock price movements on the trading day immediately following these calls. To facilitate this research, we developed the MiMIC (Multi-Modal Indian Earnings Calls) dataset, encompassing companies representing the Nifty 50, Nifty MidCap 50, and Nifty Small 50 indices...

[83] [A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist](#)

Wentao Zhang, ..., and Bo An. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2024. 102 citations.

0% Topic Match

Abstract: Financial trading is a crucial component of the markets, informed by a multimodal information landscape encompassing news, prices, and Kline charts, and encompasses diverse tasks such as quantitative trading and high-frequency trading with various assets. While advanced AI techniques like deep learning and reinforcement learning are extensively utilized in finance, their application in financial trading tasks often faces challenges due to inadequate handling of multimodal data and limited generalizability across various tasks. To address these challenges, we present FinAgent, a multimodal foundational agent with tool augmentation for financial trading. FinAgent's market intelligence module processes a diverse range of data-numerical, textual,...

[84] [RiskAwareTNet: A Multi-Task Deep Learning Framework for Stock Price Prediction and Risk Assessment](#)

Afsana Alam Nova, ..., and M. F. Mridha. 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), 2025. 0 citations.

0% Topic Match

Abstract: This paper introduces RiskAwareTNet, a novel deep-learning model designed for multi-task learning that addresses stock price prediction and risk evaluation. The model combines Bidirectional Long Short-Term Memory (BiLSTM) networks with Transformer encoders to analyze a 15-year dataset (2008–2024) for MasterCard (MA) and Visa (V), capturing complex temporal trends inherent in financial data. The model's multi-task design allows it to simultaneously predict stock prices and assess critical financial risk metrics, including volatility and the Sharpe ratio, offering a more holistic view of financial performance. RiskAwareTNet is evaluated against traditional models like ARIMA and XGBoost. The results demonstrate its superior performance, achieving...

[85] [Exploring The Efficient Market Hypothesis for Accurate Stock Movement Prediction via Feature-Axis Transformer](#)

Kanghyeon Seo and Jihoon Yang. Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing, 2024. 1 citations.

0% Topic Match

Abstract: Stock movement forecasting is a significant challenge in financial machine-learning application fields due to its profound impact on financial markets. While the Efficient Market Hypothesis (EMH) [10] postulates that predicting stock movement is nearly impossible, recent studies using deep learning methodologies exhibit promising results. The EMH is based on the assumption that stock prices immediately incorporate all available information, such as financial statements, earnings announcements, and macroeconomic news. Input features used by the prior studies commonly include Open, High, Low, Close, and Adjusted Close (OHLC), which potentially contain salient information for forecasting movements. However, most have yet to extract the...

[86] [Earnings Releases, Anomalies, and the Behavior of Security Returns](#)

George Foster, ..., and T. Shevlin. Unknown journal, 2016. 441 citations.

0% Topic Match

No summary or abstract available

[87] [Bridging the Information Gap: Quarterly Conference Calls as a Medium for Voluntary Disclosure](#)

S. C. Tasker. Review of Accounting Studies, 1999. 383 citations.

0% Topic Match

No summary or abstract available

[88] [Mastering Stock Markets with Efficient Mixture of Diversified Trading Experts](#)

Shuo Sun, ..., and Bo An. Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023. 18 citations.

0% Topic Match

Abstract: Quantitative stock investment is a fundamental financial task that highly relies on accurate prediction of market status and profitable investment decision making. Despite recent advances in deep learning (DL) have shown stellar performance on capturing trading opportunities in the stochastic stock market, the performance of existing DL methods is unstable with sensitivity to network initialization and hyperparameter selection. One major limitation of existing works is that

investment decisions are made based on one individual neural network predictor with high uncertainty, which is inconsistent with the workflow in real-world trading firms. To tackle this limitation, we propose AlphaMix, a novel three-stage...

[89] [Extracting the Structure of Press Releases for Predicting Earnings Announcement Returns](#)

Yuntao Wu, ..., and Andreas G. Veneris. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: We examine how textual features in earnings press releases predict stock returns on earnings announcement days. Using over 138,000 press releases from 2005 to 2023, we compare traditional bag-of-words and BERT-based embeddings. We find that press release content (soft information) is as informative as earnings surprise (hard information), with FinBERT yielding the highest predictive power. Combining models enhances explanatory strength and interpretability of the content of press releases. Stock prices fully reflect the content of press releases at market open. If press releases are leaked, it offers predictive advantage. Topic analysis reveals self-serving bias in managerial narratives. Our framework supports...

[90] [Truth or Fiction: Multimodal Learning Applied to Earnings Calls](#)

Jamshed Kaikaus, ..., and Robert Brunner. 2022 IEEE International Conference on Big Data (Big Data), 2022. 4 citations.

0% Topic Match

Abstract: A significant amount of resources have been used in both academia and industry to study the impact of financial text on company perception and performance. In order to mitigate potential adverse outcomes, companies have begun to regulate word usage based on perceived sentiment, making conventional text-based analysis less reliable. To address this, we present a multimodal bidirectional Long Short-Term Memory (LSTM) framework augmented with a cross-attention fusion mechanism trained on audio and text data obtained from quarterly earnings conferences calls. The framework is applied to two tasks: financial restatement prediction and market movement prediction. We compare the proposed model against...

[91] [Corporate Event Predictions Using Large Language Models](#)

Zhaomin Xiao, ..., and Jiancheng Li. 2023 10th International Conference on Soft Computing & Machine Intelligence (ISCM), 2023. 16 citations.

0% Topic Match

Abstract: This paper offers a thorough assessment of large language models (LLMs) in the context of corporate event prediction. To achieve this, we formally establish the task of corporate event prediction, construct a novel dataset containing summaries of earning call transcripts, and conduct comprehensive experiments involving both raw and fine-tuned variants of the primary LLMs. Our experimental findings underscore the viability of automating this intricate task using LLMs and highlight the unnecessary of additional finetuning.

[92] [Pre-Finetuning with Impact Duration Awareness for Stock Movement Prediction](#)

Cheng-Chih Chiu, ..., and Hsin-Hsi Chen. Companion Proceedings of the ACM on Web Conference 2025, 2024. 1 citations.

0% Topic Match

Abstract: Understanding the duration of news events' impact on the stock market is crucial for effective time-series forecasting, yet this facet is largely overlooked in current research. This paper addresses this research gap by introducing a novel dataset, the Impact Duration Estimation Dataset (IDED), specifically designed to estimate impact duration based on investor opinions. Our research establishes that pre-finetuning language models with IDIED can enhance performance in text-based stock movement predictions. In addition, we juxtapose our proposed pre-finetuning task with sentiment analysis pre-finetuning, further affirming the significance of learning impact duration. Our findings highlight the promise of this novel research direction...

[93] [Forecasting Realized Volatility: An Automatic System Using Many Features and Many Machine Learning Algorithms](#)

S. Li and Yushan Tang. Capital Markets: Asset Pricing & Valuation eJournal, 2021. 7 citations.

0% Topic Match

Abstract: We propose a machine learning-based automatic system for forecasting realized volatility. The system consists of two components: feature engineering and learning algorithm fitting. The system is automatic in that it requires little intervention in terms of feature and model selection. Using this system, we demonstrate large gains based on 118 features with five machine learning algorithms to forecast realized volatility for 173 stocks spanning two decades. Both sparse linear models (LASSO, PCR) and nonlinear algorithms (RF, GBRT, NN) can improve over the commonly used ordinary least squares models. High-frequency based realized features from existing risk models (HAR, MIDAS, SHAR, HARQ-F...

[94] [Cryptocurrency Bubble Detection: A New Stock Market Dataset, Financial Task & Hyperbolic Models](#)

Ramit Sawhney, ..., and S. Chava. Unknown journal, 2022. 23 citations.

0% Topic Match

Abstract: The rapid spread of information over social media influences quantitative trading and investments. The growing popularity of speculative trading of highly volatile assets such as cryptocurrencies and meme stocks presents a fresh challenge in the financial realm. Investigating such "bubbles" - periods of sudden anomalous behavior of markets are critical in better understanding investor behavior and market dynamics. However, high volatility coupled with massive volumes of chaotic social media texts, especially for underexplored assets like cryptocurrencies pose a challenge to existing methods. Taking the first step towards NLP for cryptocurrencies, we present and publicly release CryptoBubbles, a novel multi-span...

[95] [Quantitative Stock Investment by Routing Uncertainty-Aware Trading Experts: A Multi-Task Learning Approach](#)

Shuo Sun, ..., and Bo An. ArXiv, 2022. 2 citations.

0% Topic Match

Abstract: Quantitative investment is a fundamental financial task that highly relies on accurate stock prediction and profitable investment decision making. Despite recent advances in deep learning (DL) have shown stellar performance on capturing trading opportunities in the stochastic stock market, we observe that the performance of existing DL methods is sensitive to random seeds and network initialization. To design more profitable DL methods, we analyze this phenomenon and find two major limitations of existing works. First, there is a noticeable gap between accurate financial predictions and profitable investment strategies. Second, investment decisions are made based on only one individual predictor without...

[96] [Towards Earnings Call and Stock Price Movement](#)

Zhiqiang Ma, ..., and Xiaomo Liu. ArXiv, 2020. 4 citations.

0% Topic Match

Abstract: Earnings calls are hosted by management of public companies to discuss the company's financial performance with analysts and investors. Information disclosed during an earnings call is an essential source of data for analysts and investors to make investment decisions. Thus, we leverage earnings call transcripts to predict future stock price dynamics. We propose to model the language in transcripts using a deep learning framework, where an attention mechanism is applied to encode the text data into vectors for the discriminative network classifier to predict stock price movements. Our empirical experiments show that the proposed model is superior to the traditional...

[97] [Forecasting leading economic indicators in the US from financial news using multi-task learning](#)

Josh Jia-Ching Ying, ..., and Ji Zhang. Soft Computing, 2025. 0 citations.

0% Topic Match

No summary or abstract available

[98] [An Exploratory Study of Stock Price Movements from Earnings Calls](#)

Sourav Medya, ..., and Brian Uzzi. Companion Proceedings of the Web Conference 2022, 2022. 24 citations.

0% Topic Match

Abstract: Financial market analysis has focused primarily on extracting signals from accounting, stock price, and other numerical "hard" data reported in P&L statements or earnings per share reports. Yet, it is well-known that decision-makers routinely use "soft" text-based documents that interpret the hard data they narrate. Recent advances in computational methods for analyzing unstructured and soft text-based data at scale offer possibilities for understanding financial market behavior that could improve investments and market equity. A critical and ubiquitous form of soft data are earnings calls. Earnings calls are periodic (often quarterly) statements usually by CEOs who attempt to influence investors' expectations...

[99] [The Effect of Conference Calls on Analyst and Market Underreaction to Earnings Announcements](#)

Michael D. Kimbrough. Accounting review: A quarterly journal of the American Accounting Association, 2005. 240 citations.

0% Topic Match

No summary or abstract available

[100] [Interpretable Operational Risk Classification with Semi-Supervised Variational Autoencoder](#)

Fan Zhou, ..., and Yi Yang. Unknown journal, 2020. 8 citations.

0% Topic Match

Abstract: Operational risk management is one of the biggest challenges nowadays faced by financial institutions. There are several major challenges of building a text classification system for automatic operational risk prediction, including imbalanced labeled/unlabeled data and lacking interpretability. To tackle these challenges, we present a semi-supervised text classification framework that integrates multi-head attention mechanism with Semi-supervised variational inference for Operational Risk Classification (SemiORC). We empirically evaluate the framework on a real-world dataset. The results demonstrate that our method can better utilize unlabeled data and learn visually interpretable document representations. SemiORC also outperforms other baseline methods on operational risk classification.

[101] [DiffsFormer: A Diffusion Transformer on Stock Factor Augmentation](#)

Yuan Gao, ..., and Bolin Ding. ArXiv, 2024. 7 citations.

0% Topic Match

Abstract: Machine learning models have demonstrated remarkable efficacy and efficiency in a wide range of stock forecasting tasks. However, the inherent challenges of data scarcity, including low signal-to-noise ratio (SNR) and data homogeneity, pose significant obstacles to accurate forecasting. To address this issue, we propose a novel approach that utilizes artificial intelligence-generated samples (AIGS) to enhance the training procedures. In our work, we introduce the Diffusion Model to generate stock factors with Transformer architecture (DiffsFormer). DiffsFormer is initially trained on a large-scale source domain, incorporating conditional guidance so as to capture global joint distribution. When presented with a specific downstream task,...

[102] [STRUX: An LLM for Decision-Making with Structured Explanations](#)

Yiming Lu, ..., and Fei Liu. ArXiv, 2024. 3 citations.

0% Topic Match

Abstract: Countless decisions shape our daily lives, and it is paramount to understand the how and why behind these choices. In this paper, we introduce a new LLM decision-making framework called STRUX, which enhances LLM decision-making by providing structured explanations. These include favorable and adverse facts related to the decision, along with their respective strengths. STRUX begins by distilling lengthy information into a concise table of key facts. It then employs a series of self-reflection steps to determine which of these facts are pivotal, categorizing them as either favorable or adverse in relation to a specific decision. Lastly, we fine-tune an...

[103] [Multi-task Sequence to Sequence Learning](#)

Minh-Thang Luong, ..., and Lukasz Kaiser. CoRR, 2015. 823 citations.

0% Topic Match

Abstract: Sequence to sequence learning has recently emerged as a new paradigm in supervised learning. To date, most of its applications focused on only one task and not much work explored this framework for multiple tasks. This paper examines three multi-task learning (MTL) settings for sequence to sequence models: (a) the one-to-many setting - where the encoder is shared between several tasks such as machine translation and syntactic parsing, (b) the many-to-one setting - useful when only the decoder can be shared, as in the case of translation and image caption generation, and (c) the many-to-many setting - where multiple encoders...

[104] [Interdiscursivity in corporate financial communication: an analysis of earnings videos](#)

K. Rajandran. Corporate Communications: An International Journal, 2020. 5 citations.

0% Topic Match

Abstract: PurposeFinancial communication produces various texts, among which are earnings videos. The videos employ language and image in multimodal discourses to convey specific social meanings about corporate performance. The purpose of this paper is to select earnings videos and study their incorporated genres, styles and discourses.Design/methodology/approachInterdiscursivity permits hybridity because it mixes the choice of genres, styles or discourses. An interdiscursive analysis is conducted on earnings videos in English, French and Spanish from corporations in the global finance industry. It involved three sequential stages: (1) to detect the discourses, (2) to name the discourses and (3) to consider the function of the...

[105] [Volatility Prediction using Financial Disclosures Sentiments with Word Embedding-based IR Models](#)

Navid Rekabsaz, ..., and A. Hanbury. ArXiv, 2017. 69 citations.

0% Topic Match

Abstract: Volatility prediction—an essential concept in financial markets—has recently been addressed using sentiment analysis methods. We investigate the sentiment of annual disclosures of companies in stock markets to forecast volatility. We specifically explore the use of recent Information Retrieval (IR) term weighting models that are effectively extended by related terms using word embeddings. In parallel to textual information, factual market data have been widely used as the mainstream approach to forecast market risk. We therefore study different fusion methods to combine text and market data resources. Our word embedding-based approach significantly outperforms state-of-the-art methods. In addition, we investigate the characteristics of...

[106] [Generating Multiple Questions from Presentation Transcripts: A Pilot Study on Earnings Conference Calls](#)

Y. Juan, ..., and Hsin-Hsi Chen. Unknown journal, 2023. 4 citations.

0% Topic Match

Abstract: In various scenarios, such as conference oral presentations, company managers' talks, and politicians' speeches, individuals often contemplate the potential questions that may arise from their presentations. This common practice prompts the research question addressed in this study: to what extent can models generate multiple questions based on a given presentation transcript? To investigate this, we conduct pilot explorations using earnings conference call transcripts, which serve as regular meetings between professional investors and company managers. We experiment with different task settings and methods and evaluate the results from various perspectives. Our findings highlight that incorporating key points retrieval techniques enhances the...

[107] [Distilling Analysis from Generative Models for Investment Decisions](#)

Chung-Chi Chen, ..., and Yusuke Miyao. ArXiv, 2024. 0 citations.

0% Topic Match

Abstract: Professionals' decisions are the focus of every field. For example, politicians' decisions will influence the future of the country, and stock analysts' decisions will impact the market. Recognizing the influential role of professionals' perspectives, inclinations, and actions in shaping decision-making processes and future trends across multiple fields, we propose three tasks for modeling these decisions in the financial market. To facilitate this, we introduce a novel dataset, A3, designed to simulate professionals' decision-making processes. While we find current models present challenges in forecasting professionals' behaviors, particularly in making trading decisions, the proposed Chain-of-Decision approach demonstrates promising improvements. It integrates an...

[108] [Listen Closely: Measuring Vocal Tone in Corporate Disclosures](#)

Jonas Ewertz, ..., and Doron Reichmann. Journal of Accounting Research, 2025. 1 citations.

0% Topic Match

Abstract: We examine the usefulness of machine learning approaches for measuring vocal tone in corporate disclosures. We document a substantial mismatch between the widely adopted actor based training data underlying these approaches and speech in corporate disclosures. We find that existing models achieve near perfect vocal tone classification within their training domain. However, when tested on actual executive speech during conference calls, their performance declines to chance levels. We thus introduce FinVoc2Vec, a deep learning model that adapts to audio recordings of conference calls and classifies the vocal tone of executive speech significantly more accurately than chance. FinVoc2Vec estimates are associated with future...

[109] [Applications of deep learning in stock market prediction: recent progress](#)

Weiwei Jiang. Expert Syst. Appl., 2020. 539 citations.

0% Topic Match

No summary or abstract available

[110] [A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options](#)

S. Heston. Review of Financial Studies, 1993. 9151 citations.

0% Topic Match

No summary or abstract available

[111] [From Facts to Insights: A Study on the Generation and Evaluation of Analytical Reports for Deciphering Earnings Calls](#)

Tomas Goldsack, ..., and Chung-Chi Chen. Unknown journal, 2024. 6 citations.

0% Topic Match

Abstract: This paper explores the use of Large Language Models (LLMs) in the generation and evaluation of analytical reports derived from Earnings Calls (ECs). Addressing a current gap in research, we explore the generation of analytical reports with LLMs in a multi-agent framework, designing specialized agents that introduce diverse viewpoints and desirable topics of analysis into the report generation process. Through multiple analyses, we examine the alignment between generated and human-written reports and the impact of both individual and collective agents. Our findings suggest that the introduction of additional agents results in more insightful reports, although reports generated by human experts...

[112] [Predicting Risk from Financial Reports with Regression](#)

Shimon Kogan, ..., and Noah A. Smith. Unknown journal, 2009. 320 citations.

0% Topic Match

Abstract: We address a text regression problem: given a piece of text, predict a real-world continuous quantity associated with the text's meaning. In this work, the text is an SEC-mandated financial report published annually by a publicly-traded company, and the quantity to be predicted is volatility of stock returns, an empirical measure of financial risk. We apply well-known regression techniques to a large corpus of freely available financial reports, constructing regression models of volatility for the period following a report. Our models rival past volatility (a strong baseline) in predicting the target variable, and a single model that uses both can...

[113] [Extracting information and sentiment analysis on dialogue in financial results briefing](#)

Hiromasa Nakatsuka and Yoshiyuki Suimon. Digital Finance, 2025. 0 citations.

0% Topic Match

No summary or abstract available

[114] [Modeling financial uncertainty with multivariate temporal entropy-based curriculums](#)

Ramit Sawhney, ..., and R. Shah. Unknown journal, 2021. 3 citations.

0% Topic Match

No summary or abstract available

[115] [Stock Market Prediction via Deep Learning Techniques: A Survey](#)

Jinan Zou, ..., and Javen Qinfeng Shi. ArXiv, 2022. 39 citations.

0% Topic Match

Abstract: Existing surveys on stock market prediction often focus on traditional machine learning methods instead of deep learning methods. This motivates us to provide a structured and comprehensive overview of the research on stock market prediction. We present four elaborated subtasks of stock market prediction and propose a novel taxonomy to summarize the state-of-the-art models based on deep neural networks. In addition, we also provide detailed statistics on the datasets and evaluation metrics commonly used in the stock market. Finally, we point out several future directions by sharing some new perspectives on stock market prediction.

[116] [Temporal Evolution of Sentiment in Earnings Calls and Its Relationship with Financial Performance](#)

Zhuxuanzi Wang, ..., and Chenyao Zhu. Applied and Computational Engineering, 2025. 5 citations.

0% Topic Match

Abstract: This study investigates the temporal evolution patterns of sentiment in earnings call transcripts and their relationship with subsequent financial performance. While existing financial sentiment analysis typically examines isolated communication events, this research adopts a longitudinal approach to detect temporal sentiment patterns across consecutive quarterly reports. We develop a multimodal analytical framework that integrates lexicon-based and deep learning methods for sentiment extraction, applied to a comprehensive dataset of 18,240 earnings call transcripts from S&P 500 companies spanning 20 consecutive quarters (2018-2022). The WMSA-Bi-LSTM architecture with multi-head self-attention mechanisms is employed to capture contextual relationships in financial discourse. Our analysis reveals distinct...

[117] [The valuation of options for alternative stochastic processes](#)

J. Cox and S. Ross. Journal of Financial Economics, 1976. 3108 citations.

0% Topic Match

No summary or abstract available

[118] [MultiFin: A Dataset for Multilingual Financial NLP](#)

R. Jørgensen, ..., and Desmond Elliott. Unknown journal, 2023. 35 citations.

0% Topic Match

Abstract: Financial information is generated and distributed across the world, resulting in a vast amount of domain-specific multilingual data. Multilingual models adapted to the financial domain would ease deployment when an organization needs to work with multiple languages on a regular basis. For the development and evaluation of such models, there is a need for multilingual financial language processing datasets. We describe MultiFin – a publicly available financial dataset consisting of real-world article headlines covering 15 languages across different writing systems and language families. The dataset consists of hierarchical label structure providing two classification tasks: multi-label and multi-class. We develop our...

[119] [The sound of confidence and doubt](#)

Xiaoming Jiang and M. Pell. Speech Commun., 2017. 106 citations.

0% Topic Match

No summary or abstract available

[120] [Numerical Claim Detection in Finance: A New Financial Dataset, Weak-Supervision Model, and Market Analysis](#)

Agam Shah, ..., and S. Chava. ArXiv, 2024. 5 citations.

0% Topic Match

Abstract: In this paper, we investigate the influence of claims in analyst reports and earnings calls on financial market returns, considering them as significant quarterly events for publicly traded companies. To facilitate a comprehensive analysis, we construct a new financial dataset for the claim detection task in the financial domain. We benchmark various language models on this dataset and propose a novel weak-supervision model that incorporates the knowledge of subject matter experts (SMEs) in the aggregation function, outperforming existing approaches. We also demonstrate the practical utility of our proposed model by constructing a novel measure of "optimism". Here, we observe the...

[121] [Exploring the Scale-Free Nature of Stock Markets: Hyperbolic Graph Learning for Algorithmic Trading](#)

Ramit Sawhney, ..., and R. Shah. Proceedings of the Web Conference 2021, 2021. 47 citations.

0% Topic Match

Abstract: Quantitative trading and investment decision making are intricate financial tasks in the ever-increasing sixty trillion dollars global stock market. Despite advances in stock forecasting, a limitation of most existing neural methods is that they treat stocks independent of each other, ignoring the valuable rich signals between related stocks' movements. Motivated by financial literature that shows stock markets and inter-stock correlations show scale-free network characteristics, we leverage domain knowledge on the Web to model inter-stock relations as a graph in four major global stock markets and formulate stock selection as a scale-free graph-based learning to rank problem. To capture the scale-free...

[122] [Volatility forecast using hybrid Neural Network models](#)

Werner D. Kristjanpoller, ..., and Marcel C. Minutolo. Expert Syst. Appl., 2014. 182 citations.

0% Topic Match

No summary or abstract available

[123] [Generalized autoregressive conditional heteroskedasticity](#)

Tim Bollerslev. Journal of Econometrics, 1986. 23191 citations.

0% Topic Match

No summary or abstract available

[124] [Text-Based Network Industries and Endogenous Product Differentiation](#)

G. Phillips and Gerard Hoberg. Journal of Political Economy, 2010. 1594 citations.

0% Topic Match

No summary or abstract available

[125] [Incorporating Stock Market Signals for Twitter Stance Detection](#)

Costanza Conforti, ..., and Nigel Collier. Unknown journal, 2022. 14 citations.

0% Topic Match

Abstract: Research in stance detection has so far focused on models which leverage purely textual input. In this paper, we investigate the integration of textual and financial signals for stance detection in the financial domain. Specifically, we propose a robust multi-task neural architecture that combines textual input with high-frequency intra-day time series from stock market prices. Moreover, we extend wt-wt, an existing stance detection dataset which collects tweets discussing Mergers and Acquisitions operations, with the relevant financial signal. Importantly, the obtained dataset aligns with Stander, an existing news stance detection dataset, thus resulting in a unique multimodal, multi-genre stance detection resource....

[126] [Detect & Describe: Deep Learning of Bank Stress in the News](#)

Samuel Rönqvist and Peter Sarlin. 2015 IEEE Symposium Series on Computational Intelligence, 2015. 12 citations.

0% Topic Match

Abstract: News is a pertinent source of information on financial risks and stress factors, which nevertheless is challenging to harness due to the sparse and unstructured nature of natural text. We propose an approach based on distributional semantics and deep learning with neural networks to model and link text to a scarce set of bank distress events. Through unsupervised training, we learn semantic vector representations of news articles as predictors of distress events. The predictive model that we learn can signal coinciding stress with an aggregated index at bank or European level, while crucially allowing for automatic extraction of text descriptions...

[127] [Deep Learning for Event-Driven Stock Prediction](#)

Xiao Ding, ..., and Junwen Duan. Unknown journal, 2015. 795 citations.

0% Topic Match

No summary or abstract available

[128] [Forecasting stock market volatility using \(non linear\) Garch models](#)

P. Franses and D. Dijk. Journal of Forecasting, 1996. 401 citations.

0% Topic Match

No summary or abstract available

[129] [Constant elasticity of variance \(CEV\) option pricing model: Integration and detailed derivation](#)

Ying-Lin Hsu, ..., and Cheng-Few Lee. Math. Comput. Simul., 2008. 62 citations.

0% Topic Match

No summary or abstract available

[130] [Risk-aware Regularization for Opinion-based Portfolio Selection](#)

Ting-Wei Hsu, ..., and Hsin-Hsi Chen. Unknown journal, 2021. 1 citations.

0% Topic Match

Abstract: Every investor faces the risk-return tradeoff when making investment decisions. Most of the investors construct a portfolio instead of putting all of their wealth on a certain stock. However, most of the previous works in the NLP community focus on predicting the movement of stocks' prices or volatilities, but do not consider the portfolio selection issue. On the other hand, few works consider unstructured data in the financial community when dealing with this issue. This paper introduces a novel opinion-based portfolio selection task, and proposes new objective functions presenting different risk appetites of investors. The empirical studies of the selecting...

[131] [Twitter mood predicts the stock market](#)

J. Bollen, ..., and Xiao-Jun Zeng. ArXiv, 2010. 5236 citations.

0% Topic Match

No summary or abstract available

[132] [Interpreting TF-IDF term weights as making relevance decisions](#)

H. Wu, ..., and K. Kwok. ACM Trans. Inf. Syst., 2008. 847 citations.

0% Topic Match

No summary or abstract available

[133] [Earnings Volatility and Earnings Predictability](#)

Ilia D. Dichev and Vicki Wei Tang. IO: Empirical Studies of Firms &amp; Markets, 2009. 606 citations.

0% Topic Match

No summary or abstract available

[134] [POST-EARNINGS-ANNOUNCEMENT DRIFT - DELAYED PRICE RESPONSE OR RISK PREMIUM](#)

V. Bernard and Jacob K. Thomas. Journal of Accounting Research, 1989. 2310 citations.

0% Topic Match

No summary or abstract available

[135] [Unlocking Financial Insights: An advanced Multimodal Summarization with Multimodal Output Framework for Financial Advisory Videos](#)

Sarmistha Das, ..., and Alka Maurya. Proceedings of the 33rd ACM International Conference on Multimedia, 2025. 0 citations.

0% Topic Match

Abstract: The dynamic propagation of social media has broadened the reach of financial advisory content through podcast videos, yet extracting insights from lengthy, multimodal segments (30-40 minutes) remains challenging. We introduce FASTER(Financial Advisory Summariser with Textual Embedded Relevant images), a modular framework that tackles three key challenges: (1) extracting modality-specific features, (2) producing optimized, concise summaries, and (3) aligning visual keyframes with associated textual points. FASTER employs BLIP-2 for semantic visual descriptions, OCR for textual patterns, and Whisper-based transcription with Speaker diarization as BOS features. A modified Direct Preference Optimization (DPO)-based loss function, equipped with BOS-specific fact-checking, ensures precision, relevance, and...

[136] [JRIRD at the NTCIR-16 FinNum-3 Task: Investigating the Effect of Numerical Representations in Manager's Claim Detection](#)

Shunsuke Onuma and Kazuma Kadowaki. NTCIR Conference on Evaluation of Information Access Technologies, 2022. 3 citations.

0% Topic Match

No summary or abstract available

[137] [Forecasting Earnings Surprises from Conference Call Transcripts](#)

Ross Koval, ..., and Xifeng Yan. Unknown journal, 2023. 16 citations.

0% Topic Match

Abstract: ,

[138] [Long Short-Term Memory](#)

Sepp Hochreiter and J. Schmidhuber. Neural Computation, 1997. 96845 citations.

0% Topic Match

No summary or abstract available

[139] [Modern Portfolio Theory, Digital Portfolio Theory and Intertemporal Portfolio Choice](#)

C. K. Jones. Econometric Modeling: Capital Markets - Portfolio Theory eJournal, 2017. 12 citations.

0% Topic Match

Abstract: The paper compares three portfolio optimization models. Modern portfolio theory (MPT) is a short-horizon volatility model. The relevant time horizon is the sampling interval. MPT is myopic and implies that investors are not concerned with long-term variance or mean-reversion. Intertemporal portfolio choice is a multiple period model that revises portfolios continuously in response to relevant signals to reduce variance of terminal wealth over the holding period. Digital portfolio theory (DPT) is a non-myopic, discrete time, long-horizon variance model that does not include volatility. DPT controls mean-reversion variances in single period solutions based on holding period and hedging and speculative demand.

[140] [Learning to Forget: Continual Prediction with LSTM](#)

Felix Alexander Gers, ..., and Fred Cummins. Neural Computation, 2000. 7065 citations.

0% Topic Match

No summary or abstract available

[141] [Hierarchical Attention Networks for Document Classification](#)

Zichao Yang, ..., and E. Hovy. Unknown journal, 2016. 4631 citations.

Not measured Topic Match

Abstract: We propose a hierarchical attention network for document classification. Our model has two distinctive characteristics: (i) it has a hierarchical structure that mirrors the hierarchical structure of documents; (ii) it has two levels of attention mechanisms applied at the word and sentence-level, enabling it to attend differentially to more and less important content when constructing the document representation. Experiments conducted on six large scale text classification tasks demonstrate that the proposed architecture outperform previous methods by a substantial margin. Visualization of the attention layers illustrates that the model selects qualitatively informative words and sentences.

[142] [Detecting Deceptive Discussions in Conference Calls](#)

D. Larcker and Anastasia A. Zakolyukina. Corporate Governance: Disclosure, 2012. 560 citations.

Not measured Topic Match

No summary or abstract available

[143] [Listening to Chaotic Whispers: A Deep Learning Framework for News-oriented Stock Trend Prediction](#)

Ziniu Hu, ..., and Tie-Yan Liu. Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining, 2017. 327 citations.

Not measured Topic Match

Abstract: Stock trend prediction plays a critical role in seeking maximized profit from the stock investment. However, precise trend prediction is very difficult since the highly volatile and non-stationary nature of the stock market. Exploding information on the Internet together with the advancing development of natural language processing and text mining techniques have enabled investors to unveil market trends and volatility from online content. Unfortunately, the quality, trustworthiness, and comprehensiveness of online content related to stock market vary drastically, and a large portion consists of the low-quality news, comments, or even rumors. To address this challenge, we imitate the learning process...

[144] [The Power of Voice: Managerial Affective States and Future Firm Performance](#)

William J. Mayew and M. Venkatachalam. S&P Global Market Intelligence Research Paper Series, 2011. 481 citations.

Not measured Topic Match

No summary or abstract available

[145] [Improving stock market prediction via heterogeneous information fusion](#)

Xi Zhang, ..., and Philip S. Yu. Knowl. Based Syst., 2017. 182 citations.

Not measured Topic Match

No summary or abstract available

[146] [Stock Embeddings Acquired from News Articles and Price History, and an Application to Portfolio Optimization](#)

Xin Du and Kumiko Tanaka-Ishii. Unknown journal, 2020. 64 citations.

Not measured Topic Match

Abstract: Previous works that integrated news articles to better process stock prices used a variety of neural networks to predict price movements. The textual and price information were both encoded in the neural network, and it is therefore difficult to apply this approach in situations other than the original framework of the notoriously hard problem of price prediction. In contrast, this paper presents a method to encode the influence of news articles through a vector representation of stocks called a stock embedding. The stock embedding is acquired with a deep learning framework using both news articles and price history. Because the...

[147] [Stock price prediction using DEEP learning algorithm and its comparison with machine learning algorithms](#)

Mahla Nikou, ..., and J. Bagherzadeh. Intell. Syst. Account. Finance Manag., 2019. 212 citations.

Not measured Topic Match

Abstract: Security indices are the main tools for evaluation of the status of financial markets. Moreover, a main part of the economy of any country is constituted of investment in stock markets. Therefore, investors could maximize the return of investment if it becomes possible to predict the future trend of stock market with appropriate methods. The nonlinearity and nonstationarity of financial series make their prediction complicated. This study seeks to evaluate the prediction power of machine learning models in a stock market. The data used in this study include the daily close price data of iShares MSCI United Kingdom exchange traded fund from...

[148] [HATS: A Hierarchical Graph Attention Network for Stock Movement Prediction](#)

Raehyun Kim, ..., and Jaewoo Kang. ArXiv, 2019. 148 citations.

Not measured Topic Match

Abstract: Many researchers both in academia and industry have long been interested in the stock market. Numerous approaches were developed to accurately predict future trends in stock prices. Recently, there has been a growing interest in utilizing graph-structured data in computer science research communities. Methods that use relational data for stock market prediction have been recently proposed, but they are still in their infancy. First, the quality of collected information from different types of relations can vary considerably. No existing work has focused on the effect of using different types of relations on stock market prediction or finding an effective way...

[149] [The best of two worlds: Forecasting high frequency volatility for cryptocurrencies and traditional currencies with Support Vector Regression](#)

Yaohao Peng, ..., and M. Montenegro. Expert Syst. Appl., 2018. 228 citations.

Not measured Topic Match

No summary or abstract available

[150] [Which Spoken Language Markers Identify Deception in High-Stakes Settings? Evidence From Earnings Conference Calls](#)

J. Burgoon, ..., and Lee Spitzley. Journal of Language and Social Psychology, 2016. 76 citations.

Not measured Topic Match

No summary or abstract available

[151] [Language Modeling for the Future of Finance: A Quantitative Survey into Metrics, Tasks, and Data Opportunities](#)

Nikita Tatarinov, ..., and S. Chava. ArXiv, 2025. 4 citations.

### Not measured Topic Match

Abstract: Recent advances in language modeling have led to a growing number of papers related to finance in top-tier Natural Language Processing (NLP) venues. To systematically examine this trend, we review 374 NLP research papers published between 2017 and 2024 across 38 conferences and workshops, with a focused analysis of 221 papers that directly address finance-related tasks. We evaluate these papers across 11 quantitative and qualitative dimensions, and our study identifies the following opportunities for NLP researchers: (i) expanding the scope of forecasting tasks; (ii) enriching evaluation with financial metrics; (iii) leveraging multilingual and crisis-period datasets; and (iv) balancing PLMs with...

- [152] [Extracting key insights from earnings call transcript via information-theoretic contrastive learning](#)

Yanlong Huang, ..., and Kunpeng Zhang. Inf. Process. Manag., 2025. 2 citations.

### Not measured Topic Match

No summary or abstract available

- [153] [Predicting bankruptcy of firms using earnings call data and transfer learning](#)

H. Siddiqui, ..., and Imran Ashraf. PeerJ Computer Science, 2023. 2 citations.

### Not measured Topic Match

Abstract: Business collapse is a common event in economies, small and big alike. A firm's health is crucial to its stakeholders like creditors, investors, partners, etc. and prediction of the upcoming financial crisis is significantly important to devise appropriate strategies to avoid business collapses. Bankruptcy prediction has been regarded as a critical topic in the world of accounting and finance. Methodologies and strategies have been investigated in the research domain for predicting company bankruptcy more promptly and accurately. Conventionally, predicting the financial risk and bankruptcy has been solely achieved using the historic financial data. CEOs also communicate verbally via press releases...

- [154] [Learning Dynamic Multimodal Implicit and Explicit Networks for Multiple Financial Tasks](#)

Gary (Ming) Ang and Ee-Peng Lim. 2022 IEEE International Conference on Big Data (Big Data), 2022. 4 citations.

### Not measured Topic Match

Abstract: Many financial forecasting deep learning works focus on the single task of predicting stock returns for trading with unimodal numerical inputs. Investment and risk management however involves multiple financial tasks - forecasts of expected returns, risks and correlations of multiple stocks in portfolios, as well as important events affecting different stocks - to support decision making. Moreover, stock returns are influenced by large volumes of non-stationary time-series information from a variety of modalities and the propagation of such information across inter-company relationship networks. Such networks could be explicit - observed co-occurrences in online news; or implicit -inferred from time-series...

- [155] [Saliency-Aware Interpolative Augmentation for Multimodal Financial Prediction](#)

Samyak Jain, ..., and Dinesh Manocha. Unknown journal, 2024. 1 citations.

### Not measured Topic Match

No summary or abstract available

- [156] [Sources and Corpora](#)

Chung-Chi Chen, ..., and Hsin-Hsi Chen. From Opinion Mining to Financial Argument Mining, 2021. 0 citations.

### Not measured Topic Match

Abstract: In this chapter, we focus on the sources of financial opinions; we group these sources by the opinion holders: insiders (Sect. 3.1), professionals (Sect. 3.2), social media users (Sect. 3.3), and journalists (Sect. 3.4). Each opinion holder may have his/her own goals when expressing opinions, resulting in different opinions from unique viewpoints.

- [157] [Numerical Claim Detection in Finance: A Weak-Supervision Approach](#)

Md. Shad Akhtar, ..., and Bryan Routledge. Journal Not Provided, 2022. 2 citations.

### Not measured Topic Match

No summary or abstract available

- [158] [Presenting an Order-Aware Multimodal Fusion Framework for Financial Advisory Summarization With an Exclusive Video Dataset](#)

Sarmistha Das, ..., and And Alka Maurya. IEEE Access, 2025. 2 citations.

### Not measured Topic Match

Abstract: Amidst the current digital era, global entrepreneurship and financial awareness dissemination has surged via online podcasts and videos showcasing insightful expertise from diverse financial domain professionals. However, existing financial summarization techniques predominantly focus on textual and numerical data, neglecting the potential of a time-saving multimodal data. Addressing this gap, we introduce FinMSG, a pioneering framework for generating concise and informative summaries from lengthy financial expert videos. Leveraging a multimodal transformer-based architecture and an ordered-aware fusion algorithm, FinMSG processes text, audio, and video features to distil key opinions and insights from diverse financial domains. Subsequently, we present FAV, a one-of-its-kind multimodal...